”ALEXANDRU-IOAN CUZA” UNIVERSITY OF IASI

FACULTY OF COMPUTER SCIENCE IASI

MASTER’S THESIS

SocialSentrix

A Web-Based System for Multi-Faceted Analysis of Online Communities

by

Andrei - Vlăduț Stan

Session: July, 2025

Advisor

Prof. Dr. Sabin - Corneliu Buraga

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Avizat,

Îndrumător lucrare de disertație,

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Abstract

Social networking has become a ubiquitous and integral part of everyday life, with billions of users spending multiple – if not several – hours a day interacting, creating, and remastering content, communities, and conversations within a multitude of social media platforms. From general-purpose networks to academic, workplace, and communication-specific spaces, the range of digital spaces guarantees that virtually every topic, interest, and social connection has its place in the online world, along with a plethora of means to access it. As social interactions continue to unfold within these digital environments, over time, people, services, and companies unknowingly build a unique and personalized image which reflects upon their values, popularity, and credibility.

These dynamic identities are not confined just to their online spaces, having impact even outside of the internet, their influence extending into and affecting the real world. From internet slang, to communication styles, and even trends often bleed into everyday language; even the rise or fall of public personalities, companies, and politicians is increasingly shaped and influenced by their online presence. Ideas born in online communities, when propelled by enough momentum, often manifest as real-world events with tangible ramifications.

This paper takes a practical approach by presenting a platform and analytical framework meant to highlight how online social environments and activities contribute to the creation of personalized digital identities. Through the use of measurable metrics, it becomes possible to extract meaningful statistics, behavioral patterns, and draw conclusions about a profile's online presence, thus, **SocialSentrix** aims to bridge the gap between online activity and its broader social resonance. By analyzing sentiment and activity on social media, the platform develops a reputation profile of a subject based on emotional tone, engagement, content, and consistency. This enables individuals, researchers, and organizations to better understand how a specific profile is perceived, how it presents itself, and what influence it may carry within its digital community.

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1. Introduction

In today's hyperconnected world, social media platforms are no longer mere means of communication and interaction; rather, they have now become worldwide public venues where personal, corporate, and political identities are constantly born, challenged, rebuilt, and destroyed. Simon Kemp (2025) claims that social media is still growing in global influence and, therefore, has become essential for personal interactions, communication, and participation within online communities [1]. These online communities go well beyond their cybernetic walls, interactions constantly shaping and redefining real-world behaviors, perceptions, and opportunities. These platforms range from professional spaces such as LinkedIn and Contra to general-purpose networks like Facebook, Reddit, and X, as well as recently developed decentralized platforms like Bluesky and Mastodon. As more and more people become users, they invest time and energy into cultivating their own online presence, the sheer amount of care and effort which these users dedicate to their digital spaces has elevated the significance of these presences. As the importance and influence of a reputation grows, so do the battles of these cybernated identities, fighting for their image, relevance, and expansion.

Additionally, online reputations, as hinted previously, affect real-world outcomes, from public images of artists and brands, to hiring decisions and academic admissions, to even public trust in institutions and electoral parties. Despite the growing impact of online identity, most users lack access to tools that would allow them to assess how their profiles are perceived across this diverse range of platforms; however, a tool addressing this need, together with the metrics and methods used for its evaluation, will be presented later in this paper.

This thesis explores the construction and analysis of digital identities through social media activities; SocialSentrix is a platform designed to examine and measure various facets of these identities' reputations. This platform aims to give users, organizations, and researchers useful insights into the dynamics of digital self-presentation through a combination of platform-based data extraction, multi-faceted metrics, and visual representations of said data and metrics.

1.1. Background and Context

Over the last two decades digital communication has matured from disordered and displaced message threads and specialized forums into cohesive, surveillanced, dynamic spaces with, slowly gaining the power to mold public debate (#BlackLivesMatter (2013-current), #MeToo (2017-current), KOSMA (2025)), drive markets (GameStop – r/WallStreetBets (2021), Influencer marketing, NFT and Crypto markets), and reconfigure the formation of communities (Reddit communities, Discord servers, Bluesky – Decentralized platforms). Social media has evolved from being a fresh twist on email and blog services, becoming now a cornerstone of in-the-moment interactions among humans, this evolution being possible due to growing internet access, the spread of smart devices, and the growth of user-generated content systems.

Modern social media platforms are not simply neutral vessels of communication but rather highly engineered systems that shape participation, identity, and attention through specially crafted designs and principles. One such principle is the "participation principle" – also known as participatory culture or the user-generated content model, which, as Jenkins (2006) describes, represents the change of media culture where users do not remain passive consumers but rather become active producers, collaborators, and commenters [2]. Platforms such as Reddit, TikTok, X, and Instagram embed these interactive mechanisms into their fundamental architecture through features like upvotes, retweets, duets, threads, remixes, likes, comments, and far more. These interactive elements are designed to maximize both user-generated content and engagement metrics, additionally, when paired with well-optimized recommendation algorithms, they serve to amplify visibility and chances for interactivity through promotional mechanisms such as feeds, hashtags, and trending lists.

According to Van Dijck (2013), such platforms rely significantly on user-generated content as their primary source of cultural and financial capital, outsourcing the content production of users and monetizing interactions through advertisements and by tracking user behaviors. This participation is gamified by making the markers of engagement visible to users, such as follower, like, and view counts (to some extent, even metrics such as Reddit Karma or LinkedIn profile statistics), which incentivize action, reaction, and construction of a public digital identity [3].

Furthermore, these affordances compel users to shape their behavior and content in order to be seen and accepted; however, due to algorithmic amplification, it ensures that emotionally charged or virality-stimulating posts receive disproportionally abundant visibility, thus creating incentives that reinforce specific forms of content and behavior over others. In doing so, participation is no longer simply an opportunity to express oneself but rather a requisite to remain relevant and to establish a meaningful reputation in this digital age.

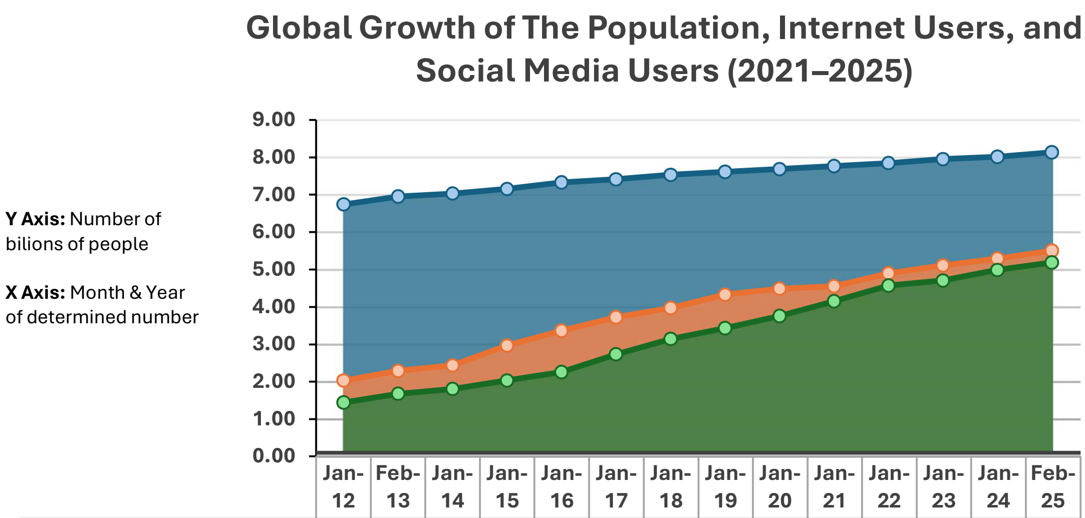
1.2. Importance of Digital Identity and Online Reputation

As specified previously, people's identities are no longer limited by physical presence, official records, or personal encounters, now extending into online spaces. Through their online behavior and interactions (posts, likes, shares, comments, following, etc), users construct public profiles which reflect their interests, values, beliefs, and affiliations. These digital personas influence how others perceive and evaluate individuals in both social and professional contexts, for example, employers, academic institutions, and even governmental institutions now consult the individual's internet activity as a first reference point for assessing their trustworthiness and character.

This dynamic is amplified by the sheer size and intensity of online participation, whereas in February 2025, there are approximately 5.24 billion active social media users – accounting for 63.9% of the world's population, with 206 million new users joining over the past year alone [1]. Based on the estimations provided by Simon Kemp (2025), we can determine that 94.24% of internet users are also social media users, with the average person now spending around 2 hours and 21 minutes per day on such platforms [1]. These estimates, along with [Figure 1.], accentuate the central role digital platforms have now taken in people's everyday life.

In comparison to our offline identity, which tends to be context-dependent and short-term, our digital identity is (usually) permanent, searchable, and algorithmically organized, based on our post history, follower count, and general metadata – all publicly/semi-publicly available. The concept of a digital identity is closely linked to the concept of online reputation, which reflects how someone is perceived and can determine certain social outcomes. This reputation is shaped not only by the nature of a user's content but also by audience reception, measured in metrics we've mentioned previously, however, also including rather abstract markers such as consistency, credibility, responsiveness, and emotional resonance.

Figure 1. Global growth of the population, internet users and social media users



	Jan-12	Feb-13	Jan-14	Jan-15	Jan-16	Jan-17	Jan-18	Jan-19	Jan-20	Jan-21	Jan-22	Jan-23	Jan-24	Feb-25
Total Population	6.8	7.01	7.09	7.21	7.39	7.47	7.59	7.67	7.75	7.83	7.91	8.01	8.08	8.2
Internet Users	2.07	2.33	2.48	3.01	3.41	3.77	4.02	4.38	4.54	4.6	4.95	5.16	5.35	5.56
Active Social Media Users	1.48	1.72	1.85	2.07	2.3	2.78	3.19	3.48	3.8	4.2	4.62	4.76	5.04	5.24

[Data determined by Simon Kemp – DataReportal.com]

1.3. Problem Statement

The ability to understand and manage one's digital identity has grown in importance as social media becomes increasingly intertwined with people's daily lives. Since more than 94% of internet users are also active social media users, using at least one, if not multiple, social platforms, these users produce enormous volumes of public activity, and yet they often lack meaningful insight into how this activity is perceived or interpreted across different platforms. While today's platforms do have certain existing analytical tools, their scope is usually limited to singular-platform metrics, being hyper-focused on engagement variables such as likes, follows, comments, views, and shares.

Even with the growing significance of digital identity, from public to professional and even institutional levels, these mono-platform tools do not provide a means to meaningfully evaluate how a user's presence is perceived across various social media platforms. Users experience both the best and worst when it comes to public visibility, sentiment, influence, and trust; yet they have little understanding of how it all comes together and how they could enhance certain areas to increase their overall digital reputation. While platforms offer isolated engagement metrics, most of these indicators lack depth and cohesion; furthermore, they lack meaningful insights into behavior patterns, consistency, and perception over time, especially when a user's presence spans across multiple spaces. Without a unified, clear, comparative framework, users are left to interpret their general reputation through fragmented feedback and anecdotal signals.

Despite the scale and reach of social media platforms, currently, there is no system which can produce a coherent, multifaceted view of one's distributed digital presence. SocialSentrix aims to address this issue by offering platform-based formulas and a tool that aggregates, analyzes, and visualizes data from multiple platforms in order to construct a richer and comprehensive picture of a user's online identity. Through the usage of the system's interactive, modular, comparable charts and its platform-specific computed scores for sentiment, engagement, trust, influence, and consistency, users can go beyond raw metrics and gain actionable insights into how their online presence is formed, maintained, and perceived. Thus, helping them to better understand their digital trajectory, draw meaningful conclusions, and make informed decisions to strengthen and refine their online identity and reputation.

1.4. Research Objectives

The primary goal of this thesis is the development of SocialSentrix, along with platform-specific formulas for computing SETIC – sentiment, engagement, trustworthiness, influence, and consistency. SocialSentrix is a web-based application designed to analyze social media profiles based on their publicly accessible, platform-specific user data, visualizing user activity, and generating a reputation score using the platform-based SETIC framework. The intended outcome of this research is to deliver a working proof of concept: a functional system that shows the feasibility of a multi-platform identity analysis through both graphical visualization and mathematical reputation score computation. This prototype will serve as groundwork for future possible expansions and refinements (such as the scoring methodology), and perhaps one day will become a potential real-world, fully-fledged application which could be of imperative use in comparative social media analytics and digital identity assessment.

1.5. Own Contributions

This dissertation presents several novel conceptual and technical contributions to the continuously evolving field of social media analysis. It primarily elaborates on **SocialSentrix** – a modular web-based visual analysis application and the **SETIC framework** – a reputation and metric calculation model, while also delving into implemented and adjacent conceptual and technological components. The SocialSentrix application enables users to extract, analyze, and visualize user data across various social media platforms – the prototype currently supports Reddit and Bluesky. Users can generate unlimited profile data graphs, in addition to the possibility of adding any number of profile datasets to their generated graphs, thus providing cross-platform and multi-profile visualization capabilities to each chart. Additionally, the visualization of the given datasets can be customized based on the selected X/Y axes elements, time frame, and interactive zooming and panning mechanisms, while also providing users with the option to export the resulting views in both image and tabular formats. Each resulting chart is fully interactive, providing tooltips and on-click (multi-)profile content rendering for all generated data points. The SETIC framework is a proposed analytical model that breaks down a profile's reputation analysis into five interconnected components (Sentiment, Engagement, Trustworthiness, Influence, Consistency). Allowing users through the web application, or direct API requests, to calculate one's overall profile reputation, in addition to individual SETIC score and component detailing. The research also dives into the conceptual understanding of identity construction, reputation modeling, and platform-based metric categorization. Elaborating on the mathematical formulas, technical concepts, platform-specific behaviors, and possible data extractions when modeling each SETIC component and its method of calculation. Moreover, the overall system has a particular focus on scalability and modularity, following the footsteps of modern revolutionary applications in dividing and allowing both technological and conceptual components to be easily accessible and modifiable. All of these technological features and conceptual foundations, together, provide a unique, practical, expandable, and interpretable approach to online reputation and digital identity assessment.

1.6. Scope and Limitations

This research is focused on the development of a proof-of-concept platform and SETIC computational formulas. While the system will allow the selection of a social media platform and the user input, the scope is currently limited due to the amount of data and requests that can be made. The system employs platform-specific formulas to compute SETIC metrics due to the variations in platform architecture, available data fields, and API access.

As a result of the previously mentioned limitations, the accuracy of direct comparisons may differ depending on the quantity and quality of the data provided by the selected platforms. Furthermore, while the system allows account login for some platforms in the scope of extracting more fields, most of the data is based on publicly available information, which limits the possibilities and precision of the formulas, thus needing to be generalized and limited.

Large-scale testing, data prediction, and other high-precision methods are currently outside the immediate scope of this prototype; however, they are acknowledged as possible future development opportunities and mentioned at the end of the thesis under chapter 6.3.

1.7. Structure of the Thesis

This thesis is organized into five main chapters, each progressively developing the context, system design, research studies, implementation, and future potential of SocialSentrix:

- ***Chapter 1: Introduction***
  Establishes the general context and background of the study, highlights the importance of digital identity and online reputation, defines the core problem and introduces a possible solution, outlines the research objectives, and provides an explanation of the thesis's scope and structure.

- ***Chapter 2: Literature Review & State of the Art***
  Reviews prior academic work on sentiment analysis, reputation systems, social media behavior modeling, and profile analytic tools. It highlights gaps in current approaches and methodologies, placing SocialSentrix within ongoing research efforts and application development.

- ***Chapter 3: Methodology***
  Elaborates upon the SETIC scoring model's logic and structure, explaining the computation process and mathematical background for how reputation is derived from activity data specific to each platform.

- ***Chapter 4: Auxiliary Social Web Architecture Concepts & Technologies***
  Explores the fundamental concepts and emerging technologies that facilitate social media interoperability and decentralized identity. Covering the AT Protocol, while

briefing over semantic graph structures, network dynamics, and micro-frontend techniques.

- ***Chapter 5: System Design & Implementation***
  Details the platform's full-stack development and system flow, including database design, routing, platform-based profile authentication, data collection logic, frontend architecture, backend infrastructure, optimization techniques, and more.

- ***Chapter 6: Conclusion & Future Work***
  Summarizes the project's results and accomplishments, mentioning known limitations, potential alternatives, and outlines future expansions.

2. Literature Review & State of the Art

The purpose of this chapter is to present an overview of existing research and technological systems on digital identity, online reputation, and social media analysis, which will lay the groundwork for the design and development of SocialSentrix in future chapters. The discussion begins with the conceptual foundations for the construction of a digital identity and the assessment of an online reputation. Followed by the exploration of existing reputation systems, sentiment analysis approaches, and other methodologies used to model public perception and behavioral patterns shown within social media. In addition, we'll study the capabilities and drawbacks of current platform-based analytic methods, as well as commercial and open-source tools used for social media monitoring and analytics. Finally, this chapter aims to reveal the conceptual and technical gaps that justify the development of SocialSentrix as a cross-platform, reputation-focused profiling solution.

2.1. Digital Identity: Concepts and Construction

A digital identity refers to the collection of data, behaviors, and representations that users create, share, or leave behind in digital spaces. In contrast to traditional identities, which form through direct interpersonal interactions, legal documents, and localized social norms and opinions, digital identities are dynamic and changing constructs, constantly being shaped by platform structures, algorithmic mediation, and online interactions. When it comes to the construction of such an identity, a complex process of multiple factors is involved, such as audience awareness, feedback loops, self-presentation, activity history, and more. While people deliberately choose what they post, like, share, or follow, the rising collection of real-world connections being digitalized forces some users to create multiple accounts or have different personas across social platforms in an attempt to limit or even hide certain likes, opinions and aspects about themselves, from spilling into their real-world image.

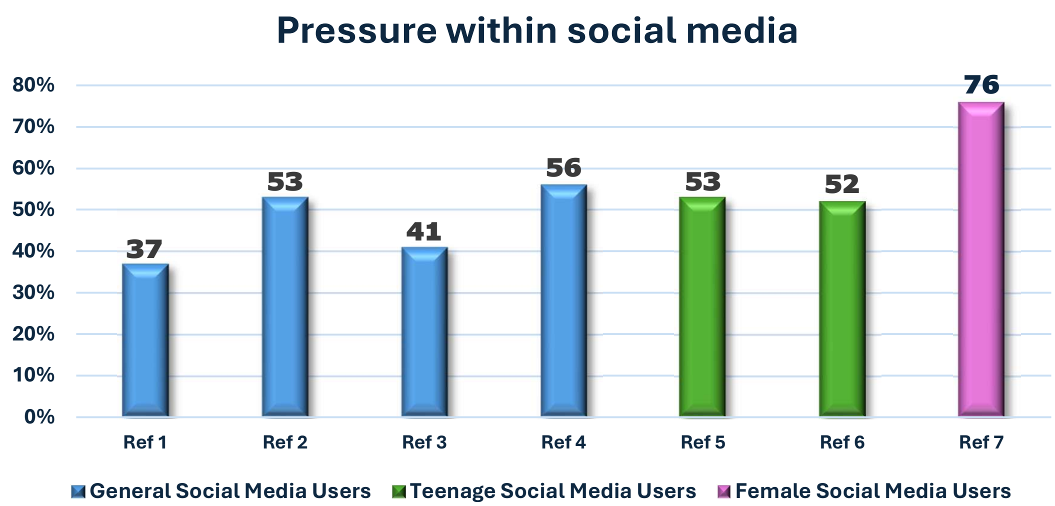
According to danah boyd's (2008) theory of "context collapse", social media complicates how people manage their online personas by bringing together a variety of audiences, where strangers, friends, family, employers, academics, and more are all brought together on a single digital stage [4].

This phenomenon was defined by Sherry Turkle (1995) as a shift towards a "multiple self" concept, in which people adopt multiple different identities in diverse online contexts, each tailored to the audiences and norms of each platform, community, and sub-community [5]. In a recent study conducted by Rosana and Fauzi (2024), after a deep dive into the process of literature-based analysis, the authors came to the conclusion and emphasized the fact that people's digital identity encompasses not only the content that they create, but also how others interpret, filter, and contextualize the respective content [6].

The article later continues and focuses on how real-time feedback mechanisms (likes, comments, shares, etc) and algorithm boosts on social media platforms enable an ongoing cycle of identity construction and adjustment, which consequently has profound implications on users' online behavior and self-perception. Due to this feedback loop, users are continuously assessing and changing their online personas to conform to the expectations of their perceived audience. People who engage in such constant self-monitoring may become more anxious and self-conscious as they look for approval and worry about judgment from their respective audiences. Such unease can eventually lead to carefully manicured online personas that value social acceptance over genuine self-expression.

While most users who express themselves more authentically are reported to have greater sense of well-being, in the study conducted by Bailey E. and Matz S. (2020), the authors discovered that most social media users – big or small are often compelled to shift towards a rather idealized self-presentation, so that they may meet certain possible social expectations. This form of self-curation can lead to a disconnection of self, alongside a general reduced welfare, thus highlighting the potential emotional and mental tolls of constant impression management in digital settings [7]. [Figure 2.] illustrates a few percentages of certain self-curation behaviors, as mentioned earlier.

Figure 2. Pressure within social media



[Data determined by Joseph D'Souza– ElectroIQ.com]

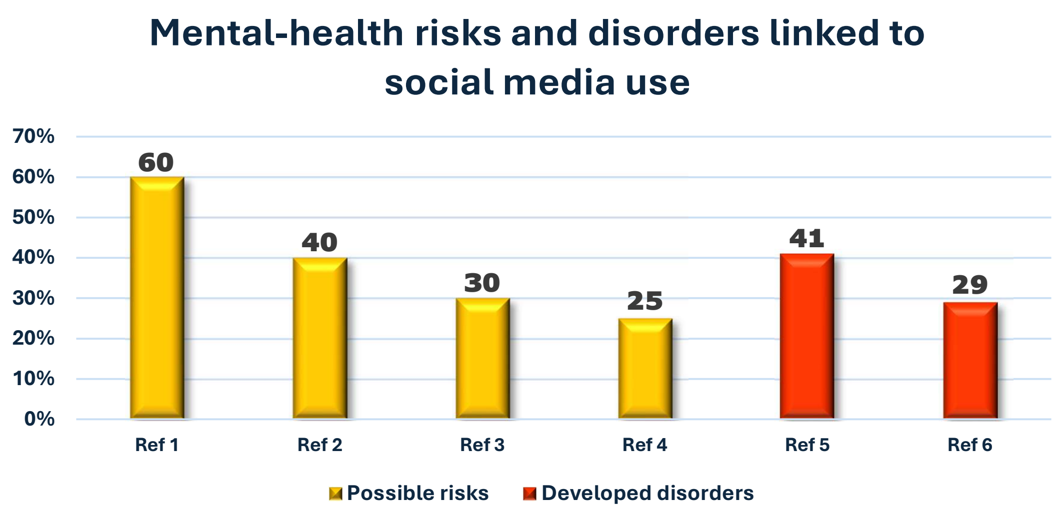
- **Reference 1:** 37% of social-media users feel pressured to create a perfect online image;
- **Reference 2:** 53% of social-media users feel upset when their posts don't get as many likes or comments as they hoped;
- **Reference 3:** 41% of social media users have removed tags from photos to avoid negative judgment;
- **Reference 4:** 56% of social-media users feel anxious when comparing themselves to their friends;

- **Reference 5:** 53% of teenagers worry about their appearance in photos posted online;
- **Reference 6:** 52% of teens have deleted posts to avoid negative feedback;

- **Reference 7:** 76% of female social-media users worry about body image from comparing themselves to others online;

Beyond the day-to-day pressures of social media, excess use can cause or exacerbate certain mental health issues. For example, excessive online exposure increases the risk of eating disorders by 2.2 times, and the usage of seven or more platforms increases the risk of anxiety by 3 times [7]. [Figure 3.] below presents more context of these elevated risks.

Figure 3. Mental-health risks and disorders linked to social media use



[Data determined by Joseph D'Souza– ElectroIQ.com]

- **Reference 1:** Using social media is linked to a 60% higher risk of developing low self-esteem;
- **Reference 2:** Social media use is associated with a 40% higher risk of developing narcissistic traits;
- **Reference 3:** Social media use is connected to a 30% higher risk of developing perfectionism;
- **Reference 4:** Social media use is connected to a 25% higher risk of developing impostor syndrome;

- **Reference 5:** 41% of social-media users report having sleep problems linked to social-media use;
- **Reference 6:** 29% of social-media users have felt depressed because of social media;

Furthermore, even though 55% of users recognize that most other users showcase an overly positive picture of their lives within these platforms, many still internalize these adverse comparisons:

- 39 % end up feeling worse about their own lives after viewing others' posts;
- 38 % of adult users say the platforms leave them feeling lonely;
- 42 % feel jealous or left out while scrolling;
- 62 % feel like their accomplishments and experiences fall short;
- 60 % report that social media harms their self-esteem;

Soh (2024) put forth a methodical framework to examine the ways in which platform affordances such as visibility settings, profile customization, or trending algorithms interact with the individual agency to shape one's identity over time [8]. Digital identities have become crucial even in policy and regulatory spheres, where governments and institutions have started to define the concept of a digital identity through legal and technical infrastructures, since having digital interactions mediate access to services more and more. The eIDAS Regulation (EU 910/2014) lays the groundwork for electronic identification and trust services in all of its member states. The mutual recognition of the validity and importance of these identities for cross-border access to public services is made possible by this regulation. This regulation illustrates just how much digital identities are growing in importance, becoming framed as a fundamental right and a public utility rather than simply a byproduct of social media usage, additionally, reflecting the growing concerns about user control, data privacy, and interoperability.

As digital identity becomes both socially performative and structurally institutionalized, more demands are arising on maintaining these images, both in terms of how one is seen and how one must appear across systems and spaces. Thus, due to social interactions and algorithmic shaping, users must be more aware of how their online personas are created and perceived while their digital identity continues to change. It's crucial to find the balance between platform-driven engagement demands (trends, content types, etc) and authenticity, especially in spaces where visibility and approval often shape behavior. In an increasingly performative digital environment, being aware of these influencing factors enables users to safeguard not only their own but also others' psychological health, in addition to presenting themselves using better techniques.

2.2. Real-world Consequences of Social Media Engagement

We can observe the importance and effects of a reputable online presence even through recent and past public events, such as political and social movements. A notable instance would be the #Blockout2024 movement, which demonstrated how celebrities, influencers, and other public figures could face large-scale reputational backlash for passive behaviors by failing to address real-life and political issues and thus failing to maintain not just a presence but ethical alignment as well. In today's digital environment, presence is no longer enough, as ideological and moral alignment is demanded as well. Papacharissi (2014) refers to this expectation as "affective publics", noting that "connective affordances of social media [...] enable expression and information sharing that liberates the individual and the collective imaginations" [9]. As Papacharissi affirms, digital affordances allow emotionally charged reactions to spread quickly and widely due to collective interpretations usually present within social networks. She also states that "these processes produce affective statements that mix fact with opinion, and with emotion, in a manner that simulates the way that we politically react in our everyday lives" [9], which directly supports the idea that digital reputation is not simply shaped by visibility, but rather is an amalgamation of nuanced elements, including platform-based ethical resonance.

The 2025 Kids Off Social Media Act (KOSMA) triggered international debate concerning potential long-term psychological and reputational risks tied to early social media exposure. This act proposes the prohibition of children under the age of 13 from accessing certain social media services, as well as the prohibition of recommendation algorithms for users under the age of 17. A proposal with deep roots in scientific research, KOSMA is supported by findings from both medical and psychological experts such as Costello et al. (2024) who stated that "social media algorithms that push extreme content to vulnerable youth are linked to an increase in mental health problems for adolescents, including poor body image, eating disorders, and suicidality" [10]. Their study not only elaborated on the concerning mental health effects these social networks have, but also on how these platforms are incentivized to continue their harmful practices due to financial gains ($11 billion alone being earned annually from the 0-17 age group) [10]. Additionally, Dr. Vivek Murthy – Office of the Surgeon General (2023) warns the population that "frequent social media use may be associated with distinct changes in the developing brain [...] and could increase sensitivity to social rewards and punishments" [11]. Collectively, the aforementioned findings alongside many more statements and research publications demonstrate the significant influence social media has on adolescent mental and emotional development, which has led and still leads to the introduction and continuous change of (new) laws, proving that these social networks can even cause research and legislative changes within the real world.

#BlackLivesMatter and #MeToo are just two other movements that have illustrated how individuals can collectively mobilize through social media to hold powerful figures and institutions accountable, pressuring companies to take public stances and revise internal policies. #BlackLivesMatter is a movement that first appeared in 2013, with its most recent and impactful resurgence being in 2020. People globally began to use social media platforms to organize protests, document police violence, and drive international discourse on racial injustice. As Yang G. (2016) highlights in his 2016 study, "the temporal unfolding of such an incident is a process of people interacting with one another and collectively creating a larger narrative […] #BlackLivesMatter becomes a unifying theme of multiple stories about racial justice.", the hashtag wasn't used simply as a digital slogan, but rather a body for organizing protests, expressing personal events, and global solidarity [12]. #MeToo also gained global momentum in 2017, empowering victims of sexual harassment and assault to share their experiences, removing the "hush topic" stigma surrounding the issue. This movement was led by widespread public support and intensified demands for justice, ultimately leading to career-altering reputational damage to several high-profile figures and businesses. As Zhang Ziqian (2023) highlights, "social media platforms provide a new way for people to discuss online and provide victims with a space to support them. And it is likely to make long-term changes in politics, culture and the world" [13]. These movements and events highlight how digital platforms can empower marginalized voices and convert silenced, private experiences into public campaigns with far-reaching consequences.

While activism is very noble and increasingly important in today's society, there are also cases in which such platforms change economic behavior and create market instability. A notable example in reference to the mentioned instability is the "GameStop Short Squeeze" of 2021, which originated from one of Reddit's communities (r/WallStreetBets), showing how these social ecosystems can affect financial markets. Due to GME (GameStop) having the most-shorted stock at the time, through the usage of memes, user reputations, and a shared revolutionary attitude, in 2021, "YOLO" buys surged, causing dramatic stock spikes. This event not only drew the attention of both economists and legislators but also became a point of study for many articles and researchers. In Desiderio et al.'s 2025 study, the authors pointed out that "increasing Reddit discussions anticipated high trading volumes […] collective investment of the community […] closely mirrored the market capitalization of the stock" [14], their earlier 2022 article describing the event as “a paramount example of collective coordination action on social media, resulting in large-scale consensus formation and significant market impact” [15]. This GME surge resulted in multibillion-dollar losses and prompted an official investigation, which, alongside researchers, revealed and elaborated on how online communities can cause substantial real-world market shifts seemingly overnight. Aside from sudden “market investments”, another component that has integrated itself as an additional economic modifier is the emergence of influencer marketing. The profitability of influencer-based sponsorships and advertisements has become increasingly obvious over the past years. Pan et Al.’s 2023 review article confirmed and elaborated on how “influencer marketing significantly impacts consumer behavior and decision-making”, which in turn led businesses to increasingly seek out and establish symbiotic relations with popular / on-theme online personalities [16]. Due to influencer marketing quickly becoming a norm, it prompted the emergence of more studies on this topic, with Liu and Zheng (2024) stating that “the informative value of influencers’ content, authenticity, and homophily positively affect their parasocial relationships”, which consequently shape brand trust and purchase behavior [17]. Similarly, the rise of NFTs (non-fungible tokens) and cryptocurrencies represents significant shifts in how digital assets are owned, valued, and traded, with projects building billion-dollar markets due to social media promotion. When an NFT or cryptocurrency is first launched, influencers promote these tokens/currencies, rapidly forming sentiment markets, using memes, trending hashtags, and viral content to often spark interest, creating sudden price surges and crashes. Lennart Ante’s 2021 research elaborates upon how certain popular online personalities such as Elon Musk, can influence crypto markets simply through social media posts. Within the article, they noted “significant positive abnormal returns and trading volume following such event”, also stating “it is reasonable to assume that his statements influence investor behavior and consequently have a market impact” [18].

From individual academic, social, and professional effects, to political movements and social activism, to influencer marketing and crypto speculation, internet presence has become a high-stakes terrain, where visibility, trustworthiness, and interaction are not just personal optional possibilities, but rather structurally valued assets. The ability to craft, sustain, and analyze one’s digital presence has become essential, as digital environments continue to expand, the ability to manage and interpret online reputation will remain, if not, actually become even more of a critical factor in navigating modern social landscapes.

2.3. Online Reputation: Theories and Impact

The term “online reputation” describes the collective perception of a person, group, or brand based on their online behavior from various online platforms. It is heavily influenced by how a user's audience engages with, comprehends, and reacts to their content rather than just what they post or produce. In contrast to digital identities, which involve self-representation, online reputation is based on how others perceive and value the respective identity. Several scholars have emphasized that online reputation is intrinsically relational and cumulative. Some of these scholars assert that reputation in social media systems is the result of the accumulation of different trust signals, including ratings, interactions, endorsements, and feedback. Credibility and reliability are established in online communities by the frequent social and algorithmic interpretation of these signals. Visible metrics such as follower counts, likes, karma scores, verified badges, and engagement ratios are frequently used to measure reputation as well; however, while these indicators do seem useful at first glance, academics have criticized them as being superficial or gamified metrics that frequently reward popularity over substance or authenticity [3]. Although more difficult to measure, subtle elements like responsiveness, emotional tone, consistency, and alignment with audience values all play a rather significant role in long-term digital reputation.

Recent studies support this complexity and the importance of nuanced metrics more and more, for instance, in a 2022 report by Mahmood et al., titled “A BERT-based Deep Learning Approach for Reputation Analysis in Social Media” the authors highlight the need for more sophisticated computational techniques to evaluate online reputation beyond simple engagement counts. The neural model they proposed could analyze not only surface-level sentiment but also semantic coherence, linguistic tone, and content relevance [19]. In addition, due to the rise of AI-based technologies in recent years, scholars have begun to investigate the significance of explainability and trust in AI-based reputation systems, proposing new methods for generating interpretable, user-aware recommendations.

Thus, emphasizing the need for a open reputation systems that not only compute scores but also provides an explanation on how and why particular profiles are ranked in a particular manner. This approach being highly relevant for tools like SocialSentrix, which aim to meaningfully visualize and justify reputation scores.

Another noteworthy contribution was made by Al-Yazidi et al. (2020), who used graph theory and natural language processing to develop a hybrid architecture integrating sentiment, trust, and influence metrics [20]. Their model successfully differentiated between different types of credibility and disinformation influence on Twitter (X), demonstrating that complex relations provide more useful information. Furthermore, reputation varies by platform and context, for example, because of their insightful comments and accumulated karma, a user might be regarded as reliable and influential on Reddit, but they might have a minimal presence on other platforms such as Instagram or X (formerly Twitter), where viral and visual content have increased significance. Users find it challenging to assess their reputation across platforms due to this fragmented and alternating landscape, particularly when feedback systems differ greatly.

The SETIC (Sentiment, Engagement, Trustworthiness, Influence, Consistency) framework, which is presented later in this thesis, aims to capture a more comprehensive and interpretable representation of online reputation in cross-platform contexts. Providing a method to analyze comparative reputation by mapping and scoring how a profile's activities are received over time using platform-specific data fields and behavioral cues.

Together, these studies collectively mark a shift toward reputation models which are highly sophisticated, data-rich, and behaviorally aware, going far beyond the usual simplistic scoring systems and thus promoting research into adaptive models that can contextualize reputation in both cross-platform and real-time scenarios. Essentially, one's online reputation has become a structurally valued asset in today's networked society, impacting everything from employment and partnerships to credibility in activism or content creation. Understanding how influence and trust emerge, evolve, and are interpreted is crucial for all users who aim to assess these factors in digital spaces.

2.4. Metrics and Modeling in Social Media Contexts

When it comes to the analysis of social media behavior and user presence, as these platforms increase in complexity and methods of interaction, the use of a complex framework for capturing different aspects of online activity becomes more and more important.

Despite the fact that traditional metrics such as engagement ratios (likes-to-followers or comments-to-views) are simplistic and unable to distinguish between resonance and controversy, or between popularity and trust, they remain widespread commons, especially in spaces such as influence marketing. Due to this inaccuracy and generalized usage, multi-dimensional modeling frameworks have become popular in recent research and applied systems. For example, in graph-based analytics, users are represented as nodes and interactions between connecting edges, allowing more complex metrics to be shown, such as centrality, clustering, and engagement diffusion. Al-Yazidi et al. (2020) proposed a hybrid model through the usage of graph theory and natural language processing (NLP), which combines trust, sentiment, and influence metrics in order to compute more accurate credibility detection on platforms like Twitter [20].

Since these metrics are vital in understanding the creation, maintenance, and assessment of digital identities and reputations, this thesis presents a classification schema for the given measurements, based on the SETIC framework – Sentiment, Engagement, Trustworthiness, Influence, and Consistency. Below, we categorize the numerous observable metrics supported by these five axes into four main categories and another three supporting ones:

- {S} Content-based metrics – Sentiment and semantic alignment
  [Which focuses on what users say or share]
  - Sentiment polarity (positive / neutral / negative)
  - Emotional tone (joy, fear, anger, sadness, surprise, love, etc.)
  - Semantic relevance (alignment with trends, topics, or audience interest)
  - Media richness (usage of images, videos, audio, and other media)
  - Originality vs Repost / Remix frequency

- {I} Network-based influence – Follower count and share reach
  [Which focuses on user position and impact within the platform's social space]
  - Follower-to-Following ratio
  - Reach and impressions (number of views and number of unique viewers)
  - Engagement spread (shares across sub-spaces / communities)
  - Clustering coefficient (close-knit vs broader audiences)
  - Bridge centrality (if it connects usually separate groups)

- {C} Behavioral consistency metrics – Posting frequency and theme consistency
  [Which focuses on user routine, predictability, and patterns]
  - Post frequency and regularity (the frequency of posts done by the user based on time)
  - Time and day consistency (the time and/or day the user usually posts)
  - Topical consistency (if they post about similar subjects / in related circles)
  - Interaction style (short replies vs long-form responses)

- {E} Surface-level engagement indicators – Likes, comments, shares, and views
  [Which focuses on immediate visible interaction statistics]
  - Like / Comment / Share ratios
  - Click-through rates (the ratio of clicks to views)
  - Profile visit count (the number of views the profile gets)
  - Post / Content save count (the number of saves / bookmarks the content receives)
  - Response ratio (the ratio of received vs sent responses)

- {T} Trust and credibility indicators - Long-term reliability and legitimacy
  [Which focuses on perceived reliability]
  - Account age
  - Verification status
  - Flag history / Rule violations / Reports
  - Endorsements (public vouches done by high-profile users)
  - External linkage (what other accounts or pages are linked)

- {S & E} Audience reaction and reception - Tone and community approval
  [Which focuses on how the public receives the content]
  - Sentiment and tone of comments
  - Ratio of support vs criticism
  - Community endorsements (if they were awarded a special community item)
  - Contextual virality (how relatable or arguable a post is)

- {T & C} Cross-platform alignment metrics - Multi-platform consistency and footprint [Which focuses on common data between multiple platforms]
  - Identity consistency (if the user has the same username across platforms)
  - Reputation parity (if the user has similar influence across platforms)
  - Topic / Content reuse (if the user reused or adapted content across platforms)
  - Posting behavior similarity (if the user posts under similar formats and timings)
  - Profile description and image alignment

This schema allows for scalable and comparative analytics across social platforms in addition to detailed reputation evaluations. For example, influence behavior on Reddit (determined by comment karma and subreddit engagement) might significantly differ from X (reposts and trending hashtags), necessitating contextualized, platform-based formulas. Additionally, emerging machine models such as transformer-based neural networks offer further modeling capabilities by tracking user activity in real time, predicting reputation trajectories, and even identifying unusual activity in real time, thus highlighting the need for models that go beyond static indicators.

To sum up, modern reputation analysis necessitates multifaceted and cross-platform viewpoints, where the SETIC-based schema provides SocialSentrix with a computational and interpretive bias, allowing it to meaningfully and quantifiably capture the complexity, nuances, and dynamics of online identity.

2.5. Sentiment and Behavioral Analysis on Social Media

Sentiment and behavioral analysis are central components in understanding digital personas and assessing reputation in online environments. Through these two analytical lenses, researchers can capture the emotional tone, intent, and communicative behavior, thus going past superficial engagement metrics, in both user content output and audience response. Together, these two analysis methods form the underlying basis for interpreting credibility, influence, and responsiveness within digital communities. Sentiment analysis, sometimes referred to as “opinion mining” or “emotion AI”, is the process of computationally identifying and categorizing the emotions expressed in a text, image, audio, or video.

Originally, this analysis was focused mainly on polarity detection (positive, neutral, negative); however, in recent developments, new methods and technologies have made it possible to classify emotions more deeply, going past simple polarities and into the categorization by sentiment, such as anger, joy, sadness, fear, surprise, disgust, and more. Some of these technologies go even further beyond, libraries such as Vader (Valence Aware Dictionary and sEntiment Reasoner), allows for the interpretation of different types of speech, such as: typical and contradictory negations, sentiment intensity through punctuation, emphasis through word-shapes, degree modifiers, sentiment-laden slang, emoticons, emojis, initialisms, acronyms, irony, sarcasm, and so on. In the context of social media, having such a large variety of use cases, this level of granularity is particularly important, due to these platforms' nature of informal and brief information and content.

A wide range of methods have been integrated into newer sentiment analysis technologies, from rule-based strategies and lexicon dictionaries (e.g., SentiWordNet) to machine learning models trained on annotated datasets. In recent years, contextual sentiment interpretation has been greatly improved sentiment analysis through transformer-based architectures and ML frameworks like BERT (Bidirectional Encoder Representations from Transformers). In their 2022 study, Mahmood et al. proposed a BERT-based deep learning model specifically tailored for reputation analysis on social media platforms by integrating semantic coherence and linguistic tone, allowing the model to differentiate between genuine sentiment and sarcasm or performative expression [19]. Through this fine-tuned model, the authors could go beyond surface-level sentiment, assessing other elements as well (such as contextual relevance), thus allowing their system to evaluate not just what was being said, but why it resonated within the specific spaces. Their model performed noticeably better than other benchmark systems, according to their comparative evaluation as seen in [Table 1.].

Models	Accuracy	BAC	F-Score
BERT (Mode 2)	0.73	0.66	0.67
BERT (Mode 1)	0.71	0.65	0.66
Peetz et al	-	0.52	0.55
SZTE-NLP	0.69	-	0.38
LIA	0.65	-	0.19
Popstar	0.64	-	0.37

Table 1. Comparison of Different Approaches in Reputation

| TP: True Positive | TN: True Negative | FP: False Positive | FN: False Negative |

| BAC = $\frac{1}{2}\left(\frac{TP}{(TP+FN)}+\frac{TN}{(TN+FP)}\right)$ | F-Score = $2\left(\frac{\frac{TP}{(TP+FP)}*\frac{TP}{(TP+FN)}}{\frac{TP}{(TP+FP)}+\frac{TP}{(TP+FN)}}\right)$ |

- Accuracy: The proportion of correct predictions ($\frac{TP+TN}{TP+TN+FP+F}$) – how often the model is right;
- BAC (Balanced Accuracy): The average of the true positive rate (sensitivity = $\frac{TP}{TP+FN}$) and the true negative rate (specificity = $\frac{TN}{TN+FP}$) – how well does the model perform across both classes (especially in imbalanced datasets).
- F-Score (F1 Score): The harmonic mean of precision and recall – how many positive predictions were correct (precision = $\frac{TP}{TP+FP}$) and how many actual positives were found (recall = $\frac{TP}{TP+FN}$).

While the main purpose of the authors' research was to simply illustrate BERT's effectiveness in reputation polarity analysis, we cannot ignore the fact that this significant outperformance across multiple configurations indicates the robustness of their proposed model and validates the utility of transformer-based architectures, highlighting their superior ability to parse semantic and tonal nuances, which are frequently present in social media spaces. Furthermore, applying further hyperparameter tuning and contextual refinement, modified versions of BERT such as RoBERTa, AlBERT, and DistilBERT, could provide even better performance [19].

Behavioral analysis instead focuses on patterns, routines, and interaction styles users display over time. Metrics such as posting frequency, temporal patterns (regular vs irregular activity), and interaction behaviors (reciprocal engagement – replies/comments, hashtag usage, trending topics, tagging of other usage, etc.) are all used by such analysis systems. These behavioral indicators can show how dedicated a user is to a platform, how frequently they contribute, and how their content changes and evolves over time. From a reputation standpoint, consistency and authenticity are often correlated with trustworthiness and influence. Users are more likely to be seen as reliable and consistent contributors if they participate regularly and constructively over time.

Furthermore, these behavioral signals are used in a myriad of other systems, such as bot detection, disinformation modeling, and influence analysis. For instance, Al-Yazidi et al. (2020) combined behavioral and sentiment features within a graph-based hybrid model to differentiate between legitimate and malicious users in Twitter (X) environments [20]. In order to identify coordinated behavior, declines in credibility, and unusual spikes in activity, the authors designed a system using a combination of natural language processing (NLP) techniques and social graph metrics (such as clustering and centrality).

Even though sentiment and behavior are usually strongly separated, when combined, they can produce deeper analytical insights. While a user whose behavior is consistent, if their sentiment varies dramatically over time, this may indicate instability and manipulation. Conversely, a consistent positive tone and regular community interactions may indicate the high social capital of a user. In systems such as SETIC, where sentiment and consistency are both essential scoring factors, this combination is particularly helpful. For example, when combining engagement metrics and feedback tone with a sentiment trajectory over time, it can indicate whether a user's public image is improving, polarizing, or declining.

Platforms such as Hootsuite, Brandwatch, and Talkwalker already incorporate sentiment scoring and behavioral dashboards into their analytical offerings. However, these tools are often aimed at brand and corporate accounts, leaving aside individual users, growing start-ups, and aspiring small organizations, which often lack this granularity or interpretive depth. This highlights the necessity of reputation-focused platforms such as SocialSentrix, which models, visualizes, and compares behavioral and emotional signals across time and (digital) space. In conclusion, compared to base engagement metrics, sentiment and behavioral analysis offer a more sophisticated understanding of online identity and reputation.

Thus, making it possible to identify narrative trends, cultivate influence, and even strategically present oneself when analyzed in a cross-platform setting. As digital environments continue to evolve, the usage and combination of affective computing and longitudinal behavior modeling will become essential in order to understand and navigate complex social dynamics online.

2.6. Existing Platforms and Tools for Profile Analytics

The rise of social media as a primary mode of communication, self-presentation, and branding has led to the emergence of a wide variety of profile analytical tools. From marketers and corporate strategists to influencers, researchers, and casual users, these systems cater to a wide range of users who want to learn more about the dynamics and results of their online presence. However, despite their extensive and widespread use, these tools vary considerably in their scope, depth, and user orientation. Hootsuite, Brandwatch, Talkwalker, Iconosquare, and Sprout Social are some of the most popular platforms for social media analytics, with business, digital marketers, and branding experts being the main clients of these platforms. Their core functionalities typically revolve around surface-level metrics such as likes, shares, comments, follower growth, and hashtag performance, while also providing dashboards for influencer benchmarking, engagement monitoring, and post scheduling.

To better understand how current tools work in practice / real-world cases, we can use leading platforms as examples and examine their specific capabilities, limitations, and results.

- Sprout Social incorporates customer relationship management (CRM) features such as conversation tracking (mentions, replies, and comments across platforms) and message tagging (custom tags in individual messages/threads), allowing brands to analyze not only what is being said, but also how their relationships with users evolve (seeing if a user has gone from an advocate to a critic, how specific followers interact with the brand's content, and if the tone and frequency in tagged conversations improve after specific interventions).

- Hootsuite centralizes account management, making it possible to track performance via engagement rates, impressions, and audience demographics across multiple and multi-platform accounts.

- Brandwatch and Talkwalker go one step further by combining Natural Language Processing (NLP) for sentiment analysis, keyword tracking, and trend monitoring across hundreds of millions of online sources (100, respectively 150 million sources, encompassing blogs, forums, videos, news, reviews, social networks, and others).

- Talkwalker uses predictive intelligence and visual analytics to improve its capacity to foresee social trends.

These commercial tools are primarily enterprise-focused, despite their usefulness, rarely providing personalized, interpretable models of individual or selective user reputation, instead focusing on campaign-level performance and engagement-driven analytics. Their evaluations frequently oversimplify the subtleties of credibility, consistency, or influence by using engagement as a stand-in base metric. Furthermore, they are frequently unable to track a user's digital persona across various ecosystems and lack cross-platform comparability, thus creating a significant knowledge gap regarding the impact, trust, and trajectory of a personal identity which can be a significant issue for other types of users, such as academic researchers, emerging professionals, growing organizations, and individual users.

To address this, open-source tools and libraries have gained popularity among academic and research communities. Libraries such as Tweepy (Twitter / X), Pushshift (Reddit), and Instaloader (Instagram) allow for the extraction of large volumes of user and post metadata. Although these libraries provide flexible access to raw data (including timestamps, comment histories, engagement numbers, and others), their meaningful processing and analysis necessitates technical expertise, as a result, they are meant for everyday social media users, but rather for computational social science researchers. And even in the case of these open-source libraries, their long-term viability is not guaranteed, being dependent on platform-specific APIs which, in time, may become deprecated, rate-limited, or even greatly altered without notice, thus restricting data access and rendering these tools obsolete.

One well-known example of such a case is Pushshift, a popular Reddit data API that saw major restrictions after Reddit's TOS (Terms of Service) changed in the spring of 2023. These modifications changed the way API data could be accessed and used by third-party tools, restricting its public availability. Even with the compromise of usage by Reddit moderators, the API turned from being a unique and highly useful tool for passionate independent users and researchers into another "effectively dead" project. Pushshift is a clear example that shows how changes to platform policies can have a direct effect on the accessibility and sustainability of even the most well-known of open-source tools.

A variety of experimental instruments have surfaced in scholarly settings, many of which use machine learning techniques to estimate influence, model the spread of false information, or detect bots. These research systems show advanced analytical potential by often combining graph-based network features, sentiment, and behavioral features, lacking user-facing features such as interactive dashboards, modular approaches, or real-world scalability, being frequently restricted to single platforms. Notably, there is a lack of accessible tools for comprehensive, reputation-focused profile analysis; the few solutions barely offer a structured model for assessing an individual's sentiment, influence, and consistency across platforms. Thus, users find it challenging to evaluate their extensive digital footprint and comprehend how their identity is interpreted in various contexts due to this lack of consistent, interpretable scoring systems.

SocialSentrix, the platform proposed in this thesis, aims to fill this gap, being designed for cross-platform analysis and individual-level insights, in contrast to current commercial and academic tools. In order to determine structured reputation scores across the five key domains of sentiment, engagement, trustworthiness, influence, and consistency (SETIC), it goes beyond simple metrics, giving users the ability to comprehend, track, and improve their online identity and reputation through modular visualizations and side-by-side profile comparisons. Aiming to democratize analytics and make social media introspection available to anyone interested in evaluating their digital presence in a meaningful way, not just marketing departments or research labs.

There are quite a few platforms and tools available for social media performance analysis, but the majority are either too technical or brand-focused for regular users, ignoring the multi-layered complexity of personal reputation in favor of superficial metrics. SocialSentrix and the SETIC framework are specifically created to address the unmet need for reputation-centric, cross-platform, and user-oriented tools.

2.7. Gaps and Limitations in Current Approaches

Despite the growing ecosystem of analytical tools developed for social media platforms, a thorough and insightful understanding of one's digital identity and reputation is still not completely possible, as these tools continue to be hindered by several key limitations.

While the previous section outlined the functional capabilities of the most popular commercial and academic tools, this section highlights the core concepts and structural flaws these tools often overlook, especially regarding reputation modeling, cross-platform analysis, and individual users.

- The absence of a consistent theoretical framework for interpreting reputation across platforms represents an initial fundamental gap. The majority of current tools equate success with engagement through the quantification of likes, views, and shares. This narrow view neglects certain aspects, such as cross-platform alignment, community perception, emotional resonance, and consistency of behavior, which are essential for determining long-term digital credibility.

- Interpretability remains another significant shortcoming in today's methods and tools. Users have to rely on external knowledge or platform-specific expertise to draw the necessary conclusions due to the resulting metrics often being uncontextualized raw numbers/charts. Few tools offer meaningful justifications for why a user is seen in a particular manner or what particular behaviors impact their wider social footprint. For non-technical users, their online reputation is often summarized as a "black box" due to the lack/limitation of transparency and explanations when it comes to their scores, metrics, reasons, and ways they are perceived.

- The lack of historical or comparative tracking is another structural drawback. The majority of tools evaluate a user's profile separately, making it impossible to compare profiles over time, between users, or between platforms. Thus limiting users' comprehension on how their reputation changes, whether their influence or trust is improving/declining, or their identity differs from others in similar communities. Without temporal or relational context, metrics lose their ability to be useful tools for reflective development and rather become static values.

- An additional aspect which is frequently overlooked is the contextual and dimensional aspect of one's identity across multiple platforms. Today's users interact with content in a variety of social environments (e.g., Reddit, Twitter/X, Instagram, Facebook, TikTok, LinkedIn, Bluesky, etc.), where each of these environments has its own interaction logic and cultural norms. Current tools are either platform-specific or don't offer enough useful methods for combining various data streams into one cohesive model. Because of this, even though it might be crucial for some in terms of networking, hiring, or brand building, plenty of users cannot evaluate their overall online presence.

- Another major gap lies in the usability and accessibility of most systems. Academic tools often call for coding expertise and domain-specific knowledge, whereas commercial tools are usually developed for enterprise clients with marketing objectives. Thus, there’s a lack of platforms built for individual reflection and self-assessment, tailored towards users who wish to understand and manage their online persona but lack the organizational infrastructure or technical know-how.

In summary, although surface-level metrics and content tracking have advanced significantly, the field still lacks easily comprehensible, reputation-focused systems that can offer users the methodologies necessary for identity management, reputation building, and self-understanding in increasingly complex digital environments. In response to these gaps, SocialSentrix introduces a more comprehensive method of social media analytics, leveraging the usage of the SETIC framework (Sentiment, Engagement, Trustworthiness, Influence, and Consistency) as a conceptual lens for understanding one’s online presence. It is intended to be comparative, modular, and interpretable, in contrast to engagement-driven dashboards or standalone sentiment trackers. Allowing users to track the evolution of their digital identity over time, across platforms, and in relation to others.

2.8. Cross-System Comparative Assessment

As social media continues to expand its availability and usage, a wide range of studies and analytical tools have surfaced to assist users in interpreting their digital reputation. While these tools have made significant advancements in metric usage, statistical indexing, trend detection, and other areas, they are limited in terms of interpretability. Regardless of the tool, there will always be alternatives presenting novel features and frameworks. SocialSentrix introduces a unique approach, focused on reputation modeling and personal identity evaluation through the usage of its SETIC framework and distinct data visualization. This subchapter presents cross-system comparisons between SocialSentrix and other existing analytical tools.

As stated in previous sections, there are several social media analysis tools, each with their own set of features and target audience, such as: Hootsuite / Sprout Social – Providing features such as campaign management, post scheduling, reply/reaction queue, basic metric visualization, and content preview, targeted for brands and marketing teams; Brandwatch / Talkwalker – Offering large-scale trend/content analysis and greater metric visualization, aimed for corporations and demographic analysis; SocialBlade – Presenting basic public-facing profile statistics, directed to general users and influencers; As well as certain academic/research-oriented tools. Unlike the tools mentioned above, SocialSentrix is dedicated to all users – individual users, researchers, influencers, small organizations, brands, large corporations, and more. It provides intuitive data interpretations, enabling all users to draw their own conclusions in regards to their overall performance; whether it be over time, in comparison to other selected platforms, or simply through a selected metric. Furthermore, it offers a unique profile data comparison and visualization system, along with the ability to calculate the reputation score of the given profile. Thus, going far beyond existing tools by not only taking into account engagement metrics, but also the content sentiment, profile trustworthiness, and overall influence and consistency.

SocialSentrix stands out through both cross-platform, multi-profile, visual data representations and individual SETIC component breakdown, offering in both cases easy-to-understand insights. Commercial tools lack platform-specific behavior modeling, profile reputation tracking, and proper cross-user comparisons. Furthermore, the application is based on a modular design, allowing it to be extensible to other platforms, and each SETIC component for each platform to be improved independently. In contrast to popular enterprise-grade services such as Hootsuite, Brandwatch, Sprout Social, and Talkwalker, which are usually expensive or inaccessible to individuals, SocialSentrix is user-first, offering all its technological and theoretical basis to any user of any type, encouraging further development, in hopes to be not only a useful tool, but a highly knowledgeable source for social media analysis and reputation calculation, branching into a multitude of topics and technologies.

2.9. Case Study and Use Case Evaluation

To demonstrate the practical utility of SocialSentrix, this subchapter presents two comparisons: one between the CEO profiles of the two available platforms and one between the dual platform profiles for the same multinational news organization. Through these comparisons and the usage of the SETIC framework, we can show how SocialSentrix makes it possible to interpret digital identities meaningfully, how social and content performance vary across platforms, and other possible ramifications. The data used in these comparisons started with each profile's initial interaction up to the timestamp: 27-06-2025 12:00 AM.

The first insightful comparison between the public profiles of *u/spez* and *@jay.bsky.team* shows how, despite being platform leaders, their overall reputation scores, while high, are not perfect. This comparison highlights how (based on the data available) there are quite a few differences when it comes to the leading profiles' audience interactions and online engagement patterns.

➢ ***u/spez*** (Reddit CEO) received an overall reputation score of 82%

- **Sentiment 72% -** Their overall content sentiment, while mostly positive, shows a noticeable variation between the types of content. Their posts exhibit a high degree of emotional positivity (94% positive), while comments head towards a more neutral/mixed tone (68% positive). The insights also point out the average content sentiment across interacted subreddits, the highest and lowest being *r/u_spez* (82%) and *r/AskReddit* (45%).

- **Engagement 96% -** Despite content infrequency, their interaction levels are quite high, with consistent upvote and comment ratios compared to the average subreddit metrics. The insights display several elements, such as the average and median interaction numbers for each subreddit (for example, the *r/u_spez* subreddit currently has 469 average upvotes and 337 average comments), as well as highlighting the profile's content diversity numbers (11 posts to 60 comments), and content duplication and deletion rates, both determined as 0%.

- **Trustworthiness 100% -** The profile is not only marked as a Reddit Admin, but the account is over 20 years old, with a verified email, Reddit Premium, 2 badges, 38 trophies, and is currently a moderator for several subreddits ranging from 87 to 304M (million) members.

- **Influence 70% -** Although the profile has a low post/comment number, the user's content reaches a wide audience, acquiring the total post karma of 179,424 and comment karma of 752,141.

- **Consistency 64% -** The user's activity is irregular, with lengthy stretches of inactivity sprinkled with sudden (active) spikes during disputes and platform updates. While the profile has only 30 out of 520 active weeks, their recent consistent activity has prevented their score from plummeting far further. With insights pointing towards a 1.45 coefficient variation within their 20:02 11/07/2015 – 21:24 25/06/2025 activity span.

- ***@jay.bsky.team*** (Bluesky CEO) received an overall reputation score of 83%
  - **Sentiment 61% -** Their overall content sentiment (though not as strong as Reddit's CEO), is generally positive, slightly tilting towards neutrality. Their posts and replies exhibit a generally similar degree of emotional positivity, being 63% and 60% respectively. However, the insights highlight that unlike *u/spez's* 71 items analyzed, *jay.bsky.team*'s sentiment score was calculated based on 3,992 items, making the generated 61% score far more reliable.
  - **Engagement 100% -** Within the Bluesky platform, the CEO's posts, replies, and reposts are widely viewed, liked, and shared, generating a high engagement. Insights display a total post/reply/repost count of 7,291 content items, which generated 3,601,595 likes, 132,345 replies, and 459,111 reposts.
  - **Trustworthiness 100% -** The profile is not only verified using one of the most trustworthy identities within the platform (*.bsky.team*), but also has a large following of 593,107 users and a small following-to-follower ratio of 1 to 156.34.
  - **Influence 80% -** Having the 29th largest follower base within the platform, their content frequently catches traction, with their average post gathering 940.56 likes/reposts. This average is in relation to their total number of posts/reposts (3,381), total likes (3,229,497 – excluding comments), and total reposts (420,827 – excluding comments). While their overall influence is much greater than that seen in the Reddit CEO's case, due to there being different metrics (followers, following, shares) and ratios based on each platform, we see only a 10% increase in score.
  - **Consistency 65% -** The user's activity might seem consistent, having 125 active weeks out of 131 during the activity span 19:23 17/11/2022 and 15:40 19/05/2025, however when following the visual datapoints provided by the web application there's a visible discrepancy, the number of posts, replies and reposts alternate erratically from one week to another (post and repost numbers jumping from 1-2 to 12-13 and then quickly back to 2-3, replies jumping from 9 to 90 and then back to 4-5). While there are indeed far more content items, the quick jumps and lengthy periods of low numbers combined with the lack of recent interactions (the past month) caused a higher coefficient variation of 2.31, and thus lowered the consistency score.

The second compelling case study involves the comparison of a user's official presence across different platforms, in this case comparing CNN's Reddit and Bluesky official social media profiles. *u/cnn* and @*cnn.com* both received decent overall reputations of 69% and 85% respectively; however, there are a few stark differences.

- Sentiment (R) 49% vs Sentiment (B) 45%: In both cases, the sentiment calculation resulted in a nearly neutral score, with the posts and comments/replies average sentiment score being 46% and 51% positive, in comparison to the other's 45% and 36% positive.
- Engagement (R) 100% vs Engagement (B) 100%: Both profiles showed strong engagement quality, one resulting in only 1 deleted item out of 2,692, along with a 1% duplication rate, and a diversity of 1,302 posts and 1,390 comments which would reach or even pass average expected subreddit metrics; While the other had much higher volume of activity with 11,317 posts/reposts/comments, gaining a total of 2,034,165 likes, 306,707 replies, and 458,089 reposts.
- Trustworthiness (R) 53% vs Trustworthiness (B) 100%: In this case, the Reddit profile had generated a lower score due to the account having only a verified email, 2 trophies, and an age of 2.42 years; Whereas the Bluesky profile was verified using a custom domain (cnn.com), along with having 585,758 followers, 38 followings, resulting a 1 to 15+ thousand following-to-follower ratio.
- Influence (R) 50% vs Influence (B) 80%: While both profiles gathered impressive numbers, 156,446 post karma and 25,467 comment karma, alongside 11,259 total posts/reposts, an average posts impact of 221.14 interactions, 2,032,205 likes, and 457,655 reposts (both excluding comments). It is clear that the Bluesky profile has far more content and influence.
- Consistency (R) 93% vs Consistency (B) 100%: Both profiles generated sub 1.0 coefficient variations, the Reddit profile being active 117 out of 126 weeks within the activity span of 20:57 30/01/2023 - 01:50 25/06/2025, resulting in a CV of 0.83; The Bluesky profile easily outdone the former, having a perfect 71 out of 71 weekly activity within the span of 15:14 22/02/2024 - 19:46 26/06/2025 resulting in a obviously lower CV of 0.23.

While the results do differ because of different content items and certain metrics, it serves as an example of how different platforms may have different perceptions of the same entity, not only in terms of public response, but also in terms of adjacent elements such as verification options, moderation activity, and content longevity. SocialSentrix captures such subtleties, offering visual models and textual insights in order to allow users to draw their own conclusions of any type, for example: on what platform should they mainly siphon their content, how they can improve their overall metrics, check if their overall reputation is improving, and so on.

The previous cases highlight key reasons that would justify why users and analysts would use SocialSentrix. From holistic performance metrics where users can asses multiple dimensions of online activity, to cross-platform profile comparisons, transparent calculations which provide multiple series of insights, to user-centric design, and even profile evolution over time. The tool highlights behavioral shifts and performance trends across any desired timespan and (platform-supported) social media profile. Thus, allowing individuals, influencers, organizations, researchers, and more to take advantage of customizable visualizations and detailed interpretations, in order to assist them in their specific scope, whether it be raising profile popularity, statistical analytics, or certain platform-based conclusions, it can be a unique and immensely useful tool in the hands of any user.

3. Methodology

This chapter outlines the research and development process applied to design, implement, and validate the SocialSentrix platform. This work uses a practical, system-oriented approach, focusing on the development of a prototype capable of performing real-time social media analysis and reputation scoring, alongside the construction of the SETIC framework. This framework is a multifaceted model that assesses digital identity through Sentiment, Engagement, Trustworthiness, Influence, and Consistency, serving as the foundation for this methodology. This methodology incorporates platform-specific approaches for data collection, metric formulation, and user interface design to guarantee contextual validity and functional reliability. Reddit and Bluesky are two social media platforms implemented into the prototype and examined by this study, each being handled with a tailored data extraction and analysis procedure that conforms with the general SETIC logic. The chapter is structured to outline the reputation model, discuss platform-specific implementations, explain the reasoning behind the approach, and go into detail about the technical tools and data handling that support the entire system. Thus, providing a methodological basis for assessing digital identities across various online environments by combining applied system design with analytical rigor.

3.1. Overview of the Methodology

This research adopts a design-based, prototype-driven methodology that combines the development of theoretical frameworks with system implementation in order to examine the feasibility of multi-platform reputation analysis. Rather than relying solely on abstract modeling or user surveys, I preferred to anchor this thesis in the development of a functioning web platform that operationalizes reputation scoring by combining multi-dimensional metrics and real-time social media data. The goal of converting conceptual models and original formulations of online reputation and digital identity into quantifiable results is what truly drives the base methodology. The system evaluates how digital personas are constructed, maintained, and perceived by gathering publicly available data from given social media platforms and applying custom reputation formulas derived from the SETIC framework. Visual aids (custom graphs) are provided alongside these computations to provide a more interpretable understanding of the user(s)'s online behavior.

The proposed approach is modular and platform-aware, recognizing that each social media platform has its own unique ecosystem of data types, interaction patterns, and visibility structures. For this reason, this study and proposed web application use platform-specific modules for Reddit and Bluesky, each of which is responsible for data extraction, feature engineering, and SETIC-specific computation. This modularity ensures the scoring models' contextual consistency and sensitivity while still aligning with a unified methodological framework. Data collection, preprocessing, reputation computation, and visualization are all distinct stages within the methodology. Each phase is linked to platform-specific elements, thus making both transparency and scalability possible. Due to the sensitivity and unpredictability of publicly accessible social media data, ethical data handling, user privacy, and platform compliance must be taken into account throughout the process.

By bridging the conceptual and practical aspects of online reputation analysis, this methodology not only validates the SETIC framework but also creates a replicable model for further research and system development in this intrinsic field of digital identity and reputation analysis.

3.2. The SETIC Reputation Model

At the core of SocialSentrix lies the SETIC reputation model, a five-dimensional framework designed to quantify and interpret digital identity using measurable indicators derived from user behavior on social media platforms. SETIC is an acronym for Sentiment, Engagement, Trustworthiness, Influence, and Consistency, where each represents a unique axis/value for assessing a user's online presence. The model was created in order to overcome the shortcomings of conventional single-metric assessments (likes, followers, or views), thus providing a multifaceted, platform-adaptable, and context-aware approach to digital reputation.

- The Sentiment component captures the emotional tone of user-generated content, comments, and interactions, encompassing both the audience's reaction and the sentiment which a user expresses in their own posts and comments/replies. Advanced sentiment analysis tools, such as lexicon-based and transformer-based models, are used to extract polarity (positive/neutral/negative), emotional categories (joy, anger, fear, etc.), and linguistic tone.

- Engagement measures the quantity and quality of interactions a user generates, such as likes, comments, reposts, replies, shares, and so on. This dimension takes into account normalized engagement rates, visibility dynamics (trending/viral potential), and ratios (likes per post or comments per follow) instead of just raw totals. Thus, by accounting for platform disparity, activity volatility, and follower inflation, this method allows for meaningful comparisons, ratios, and calculations.

- Trustworthiness focuses on indicators of legitimacy, reliability, and credibility. Account age, public endorsements, social connections, flag/report history, account verification status, external connections, and alignment with community standards are all examples of such indicators that can be taken into account to calculate one's trustworthiness.

- Influence reflects the user's relative importance and impact within their social graph or within the platform as a whole. Follower-to-following ratios, repost/reply diffusion, clustering behavior, and bridge centrality across communities are all elements that are taken into account when determining one's influence. Essentially, this dimension estimates how information from a profile is spread and whether it functions as hub, bridge, or peripheral node within its ecosystem (if the content stays within a community, or if it attracts and connects others from other communities, or perhaps others floating within the platform) by drawing on graph theory and social network analysis.

- Consistency represents the temporal and thematic coherence of a user's activity. It assesses the user's posting frequency and consistency, whether their topics remain focused or diverge, and whether stylistic or content-based patterns can be observed. Inconsistency or erratic behavior may be a sign of volatility or platform manipulation, while high consistency often aligns with perceived reliability and authenticity; however, ultra-high (near-perfect consistency) might be a sign of automation and pre-programmed content rather than original content-driven posts and interactions.

Each dimension is operationalized through platform-specific data fields and computation formulas, which are then normalized to a common 0-100 scale to facilitate cross-platform alignment and relative comparison.

The final profile reputation score is visualized using modular sub-windows and charts that present both SETIC scores and their sub-metric contributions, ensuring interpretability and depth, thus enabling researchers and users to investigate the elements that most influence a profile's perceived identity. In contrast to black-box engagement indices or single-score reputation systems, the SETIC model offers an open and expandable approach to analyzing online presence. Bridging the gap between quantitative social media metrics and qualitative reputation dynamics by focusing on how users' behavior is perceived and maintained over time and across platforms, rather than simply emphasizing a user's activity.

3.3. Platform-Specific Implementations of SETIC

Although the SETIC model offers a comprehensive framework for analyzing social media reputation, the platform context greatly affects the practical effectiveness. To accurately reflect the user's behaviors and interactions, different social media platforms provide unique data points and community dynamics, thus requiring a customized, platform-based approach. The following sections will provide brief explanations / citations based on external sources and scientific literature, in order to justify the selection and inclusion of specific elements within each SETIC calculation model component:

Sentiment – The emotional tone and stability of a user's communication is a fundamental component of social media analysis, influencing opinions about reliability, credibility, and intent. Sentiment analysis utilizes natural language processing (NLP) in order to quantify the emotional tone of user content in reputation models. According to Hutto & Gilbert (2015), sentiment patterns reveal how a user's tone is imprinted within their content and online communities. In their research, they demonstrated in both lexicon-based and machine learning approaches VADER's superior effectiveness in regards to capturing sentiment polarity and intensity in social media text, outperforming even human raters by a significant margin (0.96% vs 0.84% accuracy). The study emphasizes VADER's (Valence Aware Dictionary and sEntiment Reasoner) efficient and straightforward design, being especially well-suited (even statistically) for examining casual and brief language often found within social media networks [21]. The overall balancing of both interpretability and performance, makes this tool stand out as a reliable and user-friendly option in terms of social media sentiment analysis models.

It combines heuristics that take into consideration linguistic subtleties such as capitalization, punctuation, degree modifiers, negations, and even more (emojis, acronyms, initialisms, etc) with a sentiment lexicon. The gauge of a user's sentiment is of great importance when it comes to their reputation, since credibility and community trust are often associated with emotional consistency, constructiveness, and positive sentiment.

Engagement – User engagement is a visible indicator of social presence and relevance within a platform's ecosystem. Their visibility and involvement are crucial in the analysis of social media activity, as they serve as a key indicator of both influence and content relevance. The engagement can be measured through multiple metrics such as posting frequency, responsiveness, and interaction depth. According to Kwak et al. (2010), social media platforms can serve not only as social networks but as dynamic channels for rapid information and content distribution. Thus, a user's social value is largely determined by engagement metrics such as shares and replies, where a higher frequency of posts and replies results in greater user prominence [22]. High engagement also helps differentiate between influential / credible users from those with a limited reach. Active participation metrics (likes, comments, shares, etc.) also provide insights into the dynamics of information distribution, community connections, and general platform health. Such components are essential to reputation calculation and modeling because they indicate authenticity and ongoing community participation. Additionally, regardless of whether relational or broadcast-oriented interaction is the predominant mode of interaction, reputation models such as SETIC can be used to better capture the complex nature of engagement seen across various social media platforms by identifying these unique patterns.

Trustworthiness – A user's perceived and objective integrity are both encapsulated in their trustworthiness, which is established by their history, credentials, and the sincerity of their contributions and actions. It has been demonstrated that an account's perceived level of trustworthiness within social media platforms is greatly influenced by various factors and user-based features such as account age, verification status, profile completeness, content originality, and behavioral history. In Castillo et al.'s (2011) study on information credibility on Twitter, they proposed 4 feature sets that can be used to estimate credibility.

One such feature set is "user-based features", which examined and demonstrated how user attributes (such as registration age, number of followers / mutual followers, and activity level) can influence the credibility of content posted by that user. Stating that: "the level of credibility of information disseminated through social media can be estimated automatically. We believe that there are several factors that can be observed in the social media platform itself, and that are useful to asses information credibility." [23]. Their study proved that these features can significantly contribute to the prediction of content credibility and thus user trustworthiness. Additionally, their research aligns closely with the SETIC model's multi-dimensional approach to online reputation. In particular, it offers empirical support for the inclusion of trustworthiness as a distinct metric, reinforcing the importance of credibility analysis in reputation models. Trustworthiness is to be a critical component of digital reputation, particularly in social environments, where the quick spread of information can magnify both facts and misinformation. In addition to safeguarding the platform's information ecosystem, accurately assessing and reflecting a user's trustworthiness rewards genuine, responsible participation.

Influence – Dynamic forces that shape opinions and behaviors within social networks are the core elements of what influence is. It measures a user's capacity to reach and impact others, whether it be through their followers, content spread, leadership ability, or other factors. According to Cha et al.'s (2010) research, the concept of social media influence is more nuanced. While yes, it is closely linked to both their follower count and community engagement (such as retweets, replies, and moderator roles), by analyzing user activities and their impact, Cha and their team have discovered that certain community engagement metrics (such as shares and mentions) have proven to be better measures for determining a user's influence rather than mere follower count [24]. It is indeed true that users with large followings are not always the most effective at disseminating information, especially now in 2025, with short-form media being one of the most popular forms of media (Instagram reels, YouTube shorts, TikToks). However, it can still provide a reliable insight into the user's influence due to these profiles typically exhibiting stable ascending/descending patterns.

In contrast, the most challenging cases for determining an individual's influential status occur in instances where profiles experience sudden and significant surges in popularity, as well as in users who deliberately attempt to artificially inflate their metrics through the usage of bot accounts and frequent engagement with trending content. Thus, highlighting the idea that various forms of engagement and network behaviors must be considered when determining one's influential rating. Influence is a core element in evaluating online reputation. By emphasizing this element, reputation systems are able to pinpoint not only the most noticeable users but also those who genuinely spur significant activity and the spread of content, thus promoting more vibrant and lively online communities in exchange for offering the producer of such interactions and content a higher status.

Consistency – The regularity of posting, thematic focus, and sentiment stability are all examples of consistency, characteristics which are used by Chu et al. (2012) to distinguish real users from bots. Their study focuses on recognizing and categorizing automated Twitter accounts, differentiating between cyborgs (accounts with both bot and human activity) and simple bots. According to their research, automated accounts frequently exhibit unique activity patterns that distinguish them from real human users (such as unusually consistent posting intervals and repetitive content) [25]. Chu et al. emphasize the importance of behavioral analysis in the preservation of social media platforms' integrity, alongside the fact that inconsistent, abnormally consistent, and spam-driven behavior is often associated with lower credibility; it is concluded that authentic and consistent activity, based on natural variation, is essential for an account's reputation. A user's authenticity and dependability can be inferred from their authentic yet consistent behavior, encompassing not only the frequency of their activity but the caliber and genuineness of said activity over an extended period of time. Thus, marking consistency as another key metric in determining one's online reputation.

Even though each SETIC element is critically important on its own, they are deeply interconnected and often reinforce or moderate one another when determining a person's social media reputation. For instance, high engagement has higher value when combined with positive sentiment because hostile or irregular participation on a regular basis can damage a reputation rather than improve it. Trustworthiness also boosts the legitimacy of a user's influence due to the fact that those seen / known as trustworthy are more likely to influence others and promote positive dialogue. Similarly, influence can amplify the reputational impact of sentiment and engagement; a user with a large following can openly spread both positive and negative behaviors on a wider scale. Additionally, consistency serves as a stabilizer across the other SETIC dimensions, separating consistent, real patterns from irregular or fabricated activity and enhancing credibility and impact. Thus, as seen in [Figure 4.] and [Table 2.] these multidimensional relationships suggest that reputation in online environments is shaped by the dynamic interaction of emotional expression, activity, trust, reach, and behavioral stability rather than by a single characteristic or simple metrics.

[Figure 4.] presents a conceptual framework that demonstrates how sentiment, engagement, trustworthiness, influence, and consistency interact to shape one's overall online reputation.

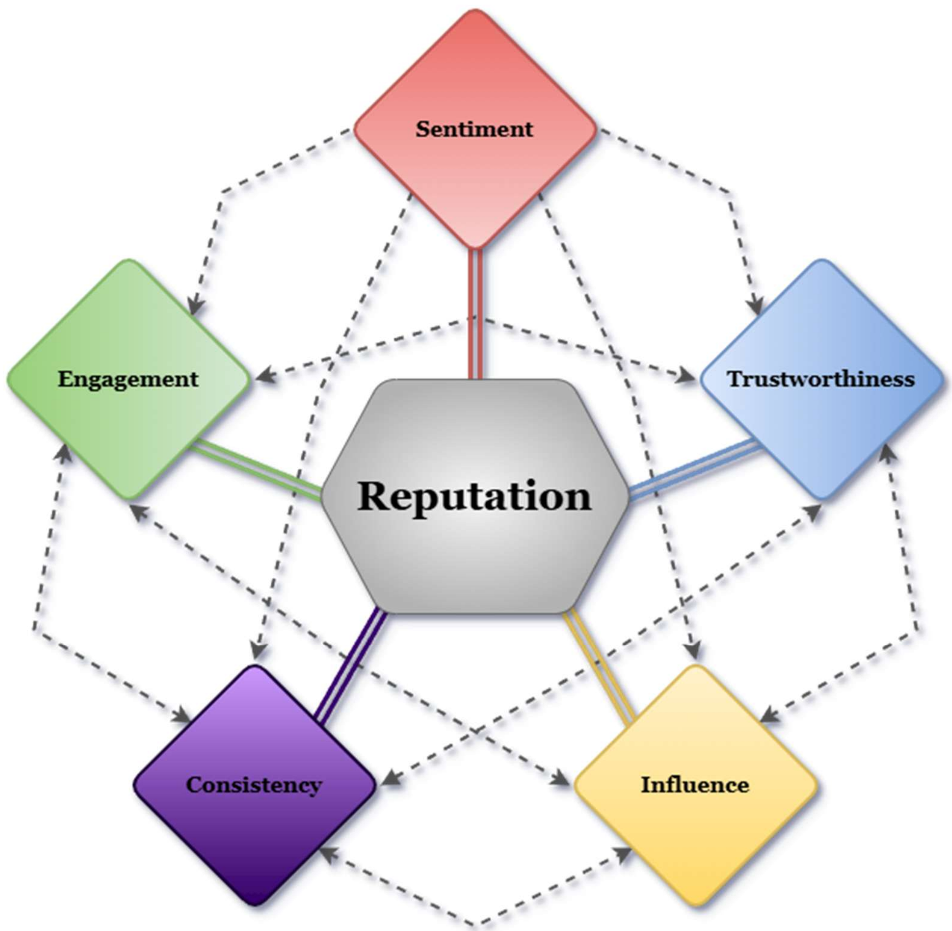


Figure 4. The SETIC Reputation Model

[Table 2.] presents an overview of the multifaceted nature of reputation calculation, showing how each SETIC model dimension influences and is influenced by the others.

From	To	Relationship/Influence Description
Sentiment	***Engagement***	Positive / Negative sentiment greatly influences how much others interact with the user's content.
Sentiment	***Trustworthiness***	Sincerity and emotional coherence are often regarded as indicators of credibility.
Sentiment	***Consistency***	Overall behavioral consistency is increased when an emotional tone is maintained over time.
Sentiment	***Influence***	Regular expression of strong feelings can directly increase a user's perceived influence.
Engagement	***Influence***	A user with high engagement can influence community conversations and reach a wider audience.
Engagement	***Trustworthiness***	Visible interactions and active participation serve as social proof, thus fostering trust.
Engagement	***Consistency***	Consistent engagement patterns reinforce user reliability and predictability.
Trustworthiness	***Influence***	Credible users frequently rise to prominence as influential figures or opinion leaders.
Trustworthiness	***Engagement***	High levels of trust attract more responses and thus, more users, leading to increased engagement.
Trustworthiness	***Consistency***	Over time, trustworthy profiles typically exhibit consistent and dependable behavior.
Influence	***Engagement***	Because of their elevated platform presence, influencers naturally receive higher levels of engagement.
Influence	***Consistency***	Influential users are frequently encouraged to continue consistent, regular activity.
Influence	***Trustworthiness***	Audiences may misinterpret influence as reliability due to authority bias.
Consistency	***Engagement***	Consistent posting practices maintain audience engagement and interest.
Consistency	***Trustworthiness***	Regular, predictable behavior fosters dependability and trust among followers and peers.
Consistency	***Influence***	Consistency over time contributes to the maintenance and expansion of a user's network influence.

Table 2. Interconnections and Pathways of Influence Between SETIC Elements

3.3.1. Reddit

Reddit is a social media platform characterized by its forum-style structure, where users of the website engage with one another in topic-based communities known as "subreddits". From news and technology to hobbies and entertainment, each subreddit focuses on a specific topic, interest, or theme. With over 3.1 million subreddits available and over 138,000 communities active daily, the plethora and diversity of these spaces ensure that any user can find their place within the platform. Reddit's unique blend of anonymity, community moderation, and open voting makes it a dynamic space for online social interaction, information sharing, and group content curation. A user's reputation is often represented by their badges, achievements, participation within communities, and their general "karma" (the net sum of their (up)/(down)votes on their posts and comments). In order to apply the SETIC framework to Reddit, these platform-specific dynamics need to be converted into quantifiable dimensions, and afterwards, based on the data the platform provides for our given user, apply certain formulas and ratios to obtain their SETIC scores and final reputation score.

Sentiment – The sentiment of a user's content is determined by simply calculating the average sentiment polarity of all their posts and comments, without taking into account the community's reaction to said contributions. The VADER NLP model is used to score the sentiment of each item (post/comment), resulting in a value ranging from -1 to 1 (absolute negative to absolute positive). After averaging these scores, the outcome is linearly mapped to a 0 to 100 scale, where 50 represents neutrality, 100 represents maximal positivity, and 0 represents maximal negativity. In the cases where users have a small/nonexistent list of content (fewer than 5 posts), their score is blended towards neutrality (50). Thus, this approach conveys the general emotional tone a user has through their Reddit activity.

Let the variables be defined as such:

N = Total number of non-empty posts and comments authored by the submitted user;

S$_i$ = Sentiment polarity score for the item *i* (determined by VADER with the range [-1,1]);

$$avg = \frac{1}{N}\sum_{i=1}^{N} S_i \qquad \boldsymbol{S_{raw}} = \begin{cases} \boldsymbol{\left(\frac{N}{5}\right) * (avg + 1) * 50 + \left(1 - \frac{N}{5}\right) * 50}, & \boldsymbol{if\ N < 5} \\ \boldsymbol{(avg + 1) * 50}, & \boldsymbol{if\ N \geq 5} \end{cases}$$

Final Sentiment score [0,100]: $\boldsymbol{S} = \max\left(0, \min(100, S_{raw})\right)$

Engagement – The user's engagement score gauges how well their posts and comments perform in relation to the interactions they receive (replies, upvotes, and downvotes), being normalized by anticipated subreddit activity. When calculating one's engagement, we use temporal weighting to prioritize recent activity, while accounting for variations in the given subreddit sizes and norms (average activity and interaction rates). This ensures fair treatment across communities regardless of their size, thus avoiding bias toward profiles and content in large and heavily trafficked communities. To further fairly represent content effort and visibility, posts are given a higher weight than comments when calculating the final score. This prioritization of original contributions over reactive ones aligns well with Cha et al.'s (2010) study, suggesting that content creation rather than content reaction leads to greater engagement and influence [24]. A weighted average is calculated after posts and comments have been independently normalized against anticipated activity levels. Additionally, to minimize noise, users with fewer than five total contributions have their score blended toward the neutral baseline of 50, thus preventing high or low scores from drastically shifting the overall component and, consequently, reputation score.

Let the variables be defined as such:

N = Total number of valid posts and comments;

P = Number of posts; C = Number of comments;

s(i) = Subreddit of item *i*; d_i = Days since item *i* was posted;

e_i = Engagement ratio of item *i*,

Where for each item *i*: $e_i = \frac{upvotes_i + comments_i}{expectedUpvotes_{s(i)} + expectedComments_{s(i)}} * exp\left(\frac{-d_i}{90}\right)$

$avgPR = avgPostRatio$
$avgCR = avgCommentRatio$

Weighted Average:
$$avgPR = \frac{1}{P}\sum_{posts} e_i$$
$$avgCR = \frac{1}{C}\sum_{comments} e_i$$

$$avgRatio = \begin{cases} 0.7 * avgPR + 0.3 * avgCR, & if\ P, C \geq 1 \\ avgPostRatio, & if\ only\ posts\ exist \\ avgCommentRatio, & if\ only\ comments\ exist \end{cases}$$

Sigmoid Transformation: $E_{sig} = \frac{100}{1+e^{-8 * (\max(avgRatio,\ 0.01) - 0.05)}}$

T = P + C = Total number of content (posts + comments);

D = Number of removed/deleted posts or comments;

R = Number of repeated posts (based on title and text);

$$duplicateRate = 1 - \frac{uniqueTitles}{P} \qquad deletionRate = \frac{D}{T}$$

$$duplicatePenalty = \begin{cases} 1 - duplicationRate, & if\ duplicateRate > 0.2 \\ 1, & otherwise \end{cases}$$

$$deletionPenalty = \begin{cases} 1 - deletionRate, & if\ deletionRate > 0.3 \\ 1, & otherwise \end{cases}$$

$$diversityBoost = \begin{cases} 1.05, & if\ duplicateRate > 0.2 \\ 1, & otherwise \end{cases}$$

$$E_{penalized} = E_{sig} * duplicatePenalty * deletionPenalty * diversityBoost$$

$$E_{raw} = \begin{cases} \left(\frac{T}{5}\right) * E_{penalized} + \left(1 - \frac{T}{5}\right) * 50, & T < 5 \\ E_{penalized}, & T \geq 5 \end{cases}$$

Final Engagement score [0,100]: $\boldsymbol{E} = \max(0, \min(100, E_{raw}))$

Trustworthiness – A user's trustworthiness is determined by a collection of visible, platform-mediated indicators that usually show reliability, genuineness, and a dedication to the platform and its rules. These indicators include verified email status, Reddit Premium (Gold) membership, account trophies, official badges (such as "Reddit Admin"), the account's tenure (age calculated by their "Cake Day"), moderation status, and total karma. Each component contributes a weighted bonus to the trustworthiness score, up to a maximum of 100 points. Since the "Reddit Admin" status (badge) signifies complete platform control and thus, trust, it takes precedence over all other considerations and automatically results in a perfect score. If not, the bonus values from all relevant indicators are bound to the [0,100] range after being added up. For example, if the user moderates several subreddits, only the largest moderated community is considered. By taking into account only the largest community, it can avoid artificial inflation from small/inactive communities, while still providing a reward for genuine large-scale moderation. Similarly, in the case of total account karma, it is assessed on a logarithmic-like scale, diminishing returns as karma increases. This approach aligns with influence modeling principles discussed by Cha et al., where in their study, they caution against linear scaling of "power-law distribution" metrics [24]. This multifactor formulation creates a consistent and interpretable trust metric for Reddit profiles by striking a balance between depth (age and karma), recognition (badges and trophies), and responsibility (moderation and verification).

Let the variables be defined as such:

$\mathbf{B_{admin}}$ = [0 or 1], based on whether the user is a Reddit Administrator;

M = Largest member count of the subreddits moderated by the user;

K = Total "karma" of the account;

A = Account age (based on current datetime and the account's "Cake Day");

V = [0 or 1], based on email verification;

P = [0 or 1], based on Reddit Premium status;

B = [0 or 1], if the number of badges (aside from admin) > 0;

T = [0 or 1], if the number of trophies > 0;

$$modBonus(M) = \begin{cases} 35, & if\ M \geq 1{,}000{,}000 \\ 15, & if\ M \geq 1{,}000 \\ 5, & if\ M \geq 1{,}000 \\ 0, & otherwise \end{cases} \qquad ageBonus(A) = \begin{cases} 35, & if\ A \geq 15 \\ 28, & if\ A \geq 10 \\ 22, & if\ A \geq 7 \\ 17, & if\ A \geq 5 \\ 12, & if\ A \geq 3 \\ 8, & if\ A \geq 2 \\ 4, & if\ A \geq 1 \\ 0, & otherwise \end{cases}$$

$$karmaBonus(K) = \begin{cases} 35, & if\ K \geq 1{,}000{,}000 \\ 25, & if\ K \geq 100{,}000 \\ 15, & if\ K \geq 10{,}000 \\ 8, & if\ K \geq 1{,}000 \\ 3, & if\ K \geq 100 \\ 0, & otherwise \end{cases} \qquad \begin{aligned} \boldsymbol{b_M} &= \mathbf{modBonus\ (M)} \\ \boldsymbol{b_K} &= \mathbf{karmaBonus\ (K)} \\ \boldsymbol{b_A} &= \mathbf{(ageBonus\ (A)} \end{aligned}$$

$$T_{raw} = \begin{cases} 100, & if\ B_{admin} = 1 \\ b_M + b_K + b_A + 15 * V + 10 * P + 5 * B + 5 * T, & otherwise \end{cases}$$

Final Trustworthiness score [0,100]: $\boldsymbol{T} = \max(0, \min(100, T_{raw}))$

Influence – The user's influence is assessed by taking into account both their structural reach (moderated audience size) and visible community approval (karma). The cumulative score of a user's contributions (karma) indicates how frequently their content is viewed favorably, calculated as a weighted sum that prioritizes original posts over comments to model proactive versus reactive contributions more accurately. By virtue of authority, users who moderate communities can increase their visibility and influence. The estimate of the user's indirect reach can be found in the total number of users in these moderated subreddits. Thus, by combining these two sources of influence (karma and audience reach), we can determine the user's final influence score, both sources being scaled to avoid extreme values from taking over. This logarithmic-based scaling compensates for extreme variations in karma averages of different subreddit sizes, considering similar principles as those being discussed by Cha et al. (2010). In the study, it was pointed out that raw popularity metrics do not correlate with actual influence, hence denoting the importance of using normalized variants [24]. The score is mapped on a 0 to 100 scale, where the higher the value, the higher the chances of their content being impactful and far-reaching.

Let the variables be defined as such:

K_p = Post karma; K_c = Comment karma;

$K = 0.75 * K_p + 0.25 * K_c$ = Weighted total karma;

M_{total} = Sum of member counts across all moderated subreddits;

b_K = Karma-based influence bonus;

b_M = Moderator audience-based influence bonus;

$$b_K = \begin{cases} 75, & if\ K \geq 1{,}000{,}000 \\ 50, & if\ K \geq 100{,}000 \\ 35, & if\ K \geq 10{,}000 \\ 25, & if\ K \geq 1{,}000 \\ 10, & if\ K \geq 100 \\ 5, & if\ K \geq 0 \\ 0, & if\ K \leq 0 \end{cases}$$

$$b_M = \min\left(20, [\log_{10}(M_{total} + 1) * 5]\right)$$

$$I_{raw} = b_K + b_M$$

Final Influence score [0,100]: $\boldsymbol{I} = \max\left(0, \min(100, I_{raw})\right)$

Consistency – The consistency of a user can be determined by looking at the regularity, rhythm, and recency of said user's activity. This score reflects how evenly a user contributes to the platform (via posts/comments) over time, by rewarding regular content posting and replies, while penalizing erratic content patterns and extended inactivity. In order to be able to measure these activity patterns, all the timestamps for posts and comments are grouped into weekly intervals. A weekly activity ratio is then calculated by dividing the total amount of time between the user's first and last activity (by the number of weeks). Additionally, weekly content count fluctuations are measured using the coefficient of variation, where a high variation denotes inconsistency. This method makes irregular posting patterns identifiable, being backed by Chu et al's (2012) study, where they used CVs to differentiate between automated and natural activity patterns [25]. Penalties are applied to reflect declining engagement if the user has not been active for a long period of time (over 3, 6, or 12 months), modeling the typical, incremental deterioration of an inactive social media profile. Thus, when taken as a whole, these components produce a score that is normalized on a scale of 0 to 100, where higher numbers indicate more recent and consistent participation over time.

Let the variables be defined as such:

$t_1, t_2, \ldots, t_n$ = Timestamps of all posts and comments authored by the user;

W_a = Number of unique weeks with at least one post/comment;

W_t = Total number of weeks between t_1 and t_n;

μ = Mean number of activities per week;

σ = Standard deviation of activities per week;

$CV = \mu / \sigma$ = (Coefficient of Variation);

M = Number of months since the user's last activity;

$$C_{raw} = 100 * \left(\frac{W_a}{W_t}\right) - \min(30, max(0, (CV - 1) * 25)) - \begin{cases} 20, & if\ M \geq 12 \\ 10, & if\ 6 \leq\ M < 12 \\ 5, & if\ 3 \leq M < 6 \\ 0, & otherwise \end{cases}$$

Final Consistency score [0,100]: $\boldsymbol{C} = \max(0, \min(100, C_{raw}))$

Final Reputation – A Reddit user's overall reputation is derived from a weighted combination of five main components: Sentiment (S), Engagement (E), Trustworthiness (T), Influence (I), and Consistency (C). Each element is calculated separately based on a specific given time period (the default being the oldest and earliest post/comment). To ensure a balanced final reputation calculation, each component has a predetermined weight (S: 20%, E: 25%, T: 20%, I: 20%, C: 15%), and once added up to the nearest integer (rounded), it results into the final reputation score R. Thus, users can monitor or compare performance across various profiles and periods due to this formulation's flexible and interpretable representation of reputation. R reflects the user's overall quality, dependability, and social impact on Reddit, through a 0 to 100 scale.

$$\boldsymbol{R = round\ (0.20 * S + 0.25 * E + 0.20 * T + 0.20 * I + 0.15 * C)}$$

3.3.2. Bluesky

Bluesky is a decentralized social media platform that aims to foster open, community-driven interactions by prioritizing algorithmic transparency and giving users control over their social connections, rather than relying on a user social graph which is centrally managed, as is the case with traditional platforms. Unlike standard centralized platforms, Bluesky profiles, posts, and engagement metrics are dispersed among independently managed servers, also known as "AT Protocol" instances. Aside from the main activities such as posting, liking, replying, and reposting, on Bluesky, users can also follow or be followed, thus creating their own networks. The quality and consistency of a user's contributions (content creation and interactions) as well as the composition of their network, determine their visibility and impact within the platform.

Thus, by having different interaction methods and other elements that could be of use, it is necessary to convert these platform-specific characteristics (follower relationships, decentralized engagement signals) into measurable and comparable reputation metrics. Through this conversion, and by ensuring that every component is calculated according to the platform's data model, we are able to apply the SETIC framework to Bluesky and its users. Only by taking into account Bluesky's specific data and logic are we able to apply certain formulas and ratios to obtain one's SETIC scores and final reputation score.

Sentiment – The sentiment of a user's content is determined based on the general tone and score of their written content (posts and replies). The VADER sentiment model is used to analyze each text (content) item, assigning a polarity between -1 (maximal negativity) and 1 (maximal positivity). By using a 0-100 scale, with 50 denoting neutrality, all of the user's content items are linearly mapped towards a general average. For stability, if the user has less than five items, their score is blended toward 50. Thus, regardless of the audience's reaction, this method yields an interpretable measure of one's expressive tone.

Let the variables be defined as such:

N = Total number of non-empty posts and comments authored by the submitted user;
S_i = Sentiment polarity score for the item *i* (determined by VADER with the range [-1,1]);

$$avg = \frac{1}{N}\sum_{i=1}^{N} S_i \qquad \boldsymbol{S_{raw}} = \begin{cases} \left(\frac{\boldsymbol{N}}{\boldsymbol{5}}\right) * (\boldsymbol{avg} + \boldsymbol{1}) * \boldsymbol{50} + \left(\boldsymbol{1} - \frac{\boldsymbol{N}}{\boldsymbol{5}}\right) * \boldsymbol{50}, & \boldsymbol{if\ N < 5} \\ (\boldsymbol{avg} + \boldsymbol{1}) * \boldsymbol{50}, & \boldsymbol{if\ N \geq 5} \end{cases}$$

Final Sentiment score [0,100]: $\boldsymbol{S} = \max(0, \min(100, S_{raw}))$

Engagement – A user's engagement score is assessed through the total number of likes, replies, and reposts received across all of the user's content during the given time period. Where the average engagement per post is calculated by dividing the total number of interactions by the number of content items. To normalize the total average based on account growth and size, it is afterwards passed through a sigmoid function with a baseline of 0.2 (typical engagement per item). This nonlinear transformation lessens the impact of outliers, preventing the existence of a few viral posts from disproportionately inflating the overall score, therefore rewarding consistent interaction. This approach way motivated by Kwak et al.'s (2010) study, due to their findings highlighting that raw follower counts frequently deviate from content influence, mentioning the need for alternative modeling techniques [22]. Mapping the outcome to a [0,100] scale produces a normalized value that enables fair comparison across users with different activity levels. However, if the user lacks any content, then the score defaults to 0.

Let the variables be defined as such:

T = Total number of content items (posts, comments, and reposts);

L | R | Q = Sum of like | reply | repost counts across all items;

$$engagementRatio = \frac{L+R+Q}{T} \quad E_{raw} = \frac{100}{1+e^{-8*(\max(engagementRatio,\ 0.01)-0.2)}}$$

Final Engagement score [0,100]: $E = round\ (E_{raw})$

Trustworthiness – A user's trustworthiness is determined through a blend of elements such as account attributes, network size, and engagement ratio. When calculating the user's trustworthiness score, it incorporates the logarithmic-scaled follower count, ratio of followers to following, and a small activity bonus (for users who have posted frequently). Through log-scaling, it limits the skew caused by follower inflation, ensuring fair resulting values for profiles with widely different follower sizes. Additionally, verified or external (non ".bsky.social") domains receive special recognition due to their perceived authenticity or institutional affiliation, thus potentially boosting their trust score. This approach was taken in reference to AT Protocol's principles, the documentation outlining that domain-linked handles are used to evidence verifiable ownership. The sum total of these elements (min 0 – capped at 100) results in the overall trustworthiness score of the given user.

Let the variables be defined as such:

F = Follower count; G = Following count; P = Number of posts;

D = [0 or 1], whether the user's handle is external (not .bsky.social)

$$followerScore = \min\ (50, round(\log_{10}(F+1)*10))$$

$$raitoScore = \min\left(30, round\left(\left(\frac{F}{G+1}\right)*10\right)\right) \quad activityBonus = \begin{cases} 100, & P \geq 10 \\ 5, & P \geq 5 \\ 0, & otherwise \end{cases}$$

$$T_{raw} = \begin{cases} 100, & D = 1\ \&\ official \\ (\ followerScore + ratioScore + activityBonus\ )*1.5, & D = 1 \\ followerScore + ratioScore + activityBonus, & D = 0 \end{cases}$$

Final Trustworthiness score [0,100]: $\boldsymbol{T} = \min(100, T_{raw})$

Influence – The user's influence is assessed by determining both the reach and the impact of their content. Thus, calculated by adding together a logarithmic follower score, a logarithmic-scaled impact score (based on average likes and reposts per item), and a bonus for viral content (which has received more than 1,000 likes). This approach uses logarithmic scaling to balance audience size and content impact by reducing the dominance effect of profiles with large followings, a common issue and solution also described by Cha et al. in their 2010 study [24]. A viral bonus highlights anomalies without letting them distort the overall impact of the content. The final influence score is capped at 100 for normalization.

Let the variables be defined as such:

F = Follower count; C = Number of content items (posts and reposts);

L = Total likes across all content; Q = Total reposts across all content;

V = Number of liked posts exceeding 1,000 likes;

$$avgImpact = \frac{L+Q}{C} \qquad \begin{array}{l} impactScore = \min\left(40, round(\log_2(avgImpact + 1) * 10)\right) \\ followerScore = \min\left(40, round(\log_{10}(F + 1) * 10)\right) \\ viralBonus = \min(10, 2 * V) \end{array}$$

$$\boldsymbol{I_{raw} = impactScore + followerScore + viralBonus}$$

Final Influence score [0,100]: $\boldsymbol{I} = \min(100, I_{raw})$

Consistency – The consistency of a user can be determined by measuring how frequently, recently, and regularly the user produces content. Every activity timestamp (posts, replies, and reposts) is grouped into weekly intervals, where the coefficient of variation is used to penalize erratic behavior. As demonstrated in Chu. et al.'s (2012) study on behavioral analysis, higher CVs suggest irregular/unusual activity patterns, whereas lower CVs highlight consistency and dependability over time. However, it is important that even in the case of low CVs, those that are extremely low (near perfect) might signal an automated or scripted behavior [25]. Bonuses are then offered for consistent participation (1-5 actions per week), while additional penalties are applied for extended periods of inactivity (greater than 3, 6, or 12 months since the last activity). Thus, resulting in a final score mapped to a [0, 100] scale.

Let the variables be defined as such:

$t_1, t_2, \dots, t_n$ = Timestamps of all posts and comments authored by the user;

W_a = Number of unique weeks with at least one post/comment;

W_t = Total number of weeks between t_1 and t_n;

μ = Mean number of activities per week; σ = Standard deviation of activities per week;

$CV = \mu / \sigma$ (Coefficient of Variation);

M = Number of months since the user's last activity;

$$C_{raw} = 100 * \left(\frac{W_a}{W_t}\right) - \min(30, max(0, (CV - 1) * 25)) - \begin{cases} 20, & if\ M \geq 12 \\ 10, & if\ 6 \leq M < 12 \\ 5, & if\ 3 \leq M < 6 \\ 0, & otherwise \end{cases}$$

Final Consistency score [0,100]: $\boldsymbol{C} = \max(0, \min(100, round(C_{raw})))$

Final Reputation – A Bluesky user's overall reputation is made up from a weighted combination of the five main SETIC components: Sentiment (S), Engagement (E), Trustworthiness (T), Influence (I), and Consistency (C). Each element is calculated separately based on a specific given time period (the default being the oldest and earliest post/reply), rounded to the nearest integer, and weighted to reflect their relative importance (S: 20%, E: 25%, T: 20%, I: 20%, C: 15%). Finally, this formulation yields a flexible and interpretable user reputation score R, allowing cross-platform and time-based comparison.

$$\boldsymbol{R = round\ (0.20 * S + 0.25 * E + 0.20 * T + 0.20 * I + 0.15 * C}$$

4. Auxiliary Social Web Architecture Concepts & Technologies

This subchapter dives into the structure and operation of contemporary social media-adjacent technology, such as fundamental protocols (AT protocol, W3C DIDs, WebID) and related technologies (SNA, FOAF, design patterns). Exploring both the theoretical and practical basis that make up both centralized and decentralized social network architectures.

4.1. Decentralized Identity – WebID and DIDs

As more and more social media platforms emerge, and previous platform increasingly expand, these platforms progressively intertwine themselves with user's personal and professional connections. Due to these platforms becoming such a staple not only in our digital identities but also our real life identity, the issue of digital identity ownership becomes increasingly prevalent. Since traditional social media networks usually lock one's sub-identity to their platform, it restricts users from being able to verify or transfer their reputation and data across different platforms. To address this issue, inspired by the recent rise of decentralized systems, the concept of decentralized identity has emerged, with there being two major initiatives: WebID and Decentralized Identifiers. These two are developed and standardized by the World Wide Web Consortium (W3C) – an international community that develops standards and guidelines for existing and new web technologies (HTML, CSS, XML, etc) meant to ensure the long-term growth of the World Wide Web.

WebID allow users to create globally unique identifiers through the usage of HTTP URIs. These URIs serve as the unique user identifier, pointing towards an RDF profile which contains user metadata, social connections (through vocabularies such as FOAF – friend of a friend), and interoperable identity graphs. Unlike WebID, DIDs take a cryptographic approach, a DID being a unique string mapped to a DID Document, which hosts the verification methods (public keys), service endpoints, and authentication details. The two digital identity models, while both decentralized, have their own use cases, with WebID being closely linked to web architecture and the semantic web stack, making it well suited for cases where interoperability and semantic meaning are core elements, while DIDs are not tied to the web and can be used in a variety of environments (blockchains, IoT devices, mobile apps, etc.) For example, a researcher can publish their WebID, linking it to certain FOAF profiles and RDF metadata about publications, allowing for the automatic traversal and indexing of the resulting linked data graphs, thus providing academic indexing and semantic search engines. On the other hand, DIDs can be used as representations for different types of data entities (e.g., a ID card, a decentralized health record, etc.). In SocialSentrix's case, adopting such a decentralized identity model could allow for far greater data access as well as in-depth user and content networking, thus boosting the accuracy of certain, if not all, SETIC components while keeping everything secure and highly portable.

4.2. Decentralized Social Architectures and the AT Protocol

Traditionally, social media platforms such as Reddit and Twitter operate within centralized systems, where their user data and interactions are managed through platform-controlled infrastructures. In contrast, Bluesky's platform embraces a decentralized approach, built on the AT Protocol (Authenticated Transfer Protocol) developed by the Bluesky project. According to Bluesky's official documentation, *atproto* is a "standard for public conversation and an open-source framework for building social apps", by creating a standardized format "for user identity, follows, and data on social apps", it enables interoperability between different applications, allowing users to move freely between platforms. The following sections describe the core concepts presented in the aforementioned documentation.

Identity:

In the AT Protocol, user identity is domain-based and decentralized, each user being identified through their unique domain name, which corresponds to a globally unique Decentralized Identifier (DID), and cryptographic URLs. For example, within the Bluesky social media platform, the official Bluesky profile (@bsky.app) is represented by the following metadata: {handle: "bsky.app", did: "did:plc:z72i7hdynmk6r22z27h6tvur", alsoKnownAs: "at://bsky.app"}. Similarly, the official SocialSentrix profile (@socialsentrix.bsky.social) is identified as: {handle: "socialsentrix.bsky.social", did: "did:plc:k5wojymvtiwypm2l6rljwree", alsoKnownAs: "at://socialsentrix.bsky.social"}.

As is noticeable in the two previous examples, this approach allows for a verifiable yet human-readable naming structure, providing users with the option of hosting their identities (DID, data repository, and social graph) independently or through specific providers such as Bluesky Social, custom domain hosts (running an independent PDS provider), and third-party hosts. This decentralized identity model, atproto, guarantees true user ownership, enabling users to maintain a consistent presence across different social networks by separating one's identity from the respective platform, thus removing the traditional platform lock-in without losing their profile's handle, content, or followers. Such functionality is especially significant in today's times, where name-tied credibility is crucial for services, organizations, and public figures, as well as the fact that the vast majority of social media users have more than just 1 platform profile.

Data Repositories:

All user-generated content is stored and managed within personal data repositories, which function as structured collections of data records, representing the user's activity (posts, replies, likes, media, follows, etc.) across the entire atproto network. Conceptually, the data repository is similar to a Git repository, where updates are versioned and secure due to each data record being signed using the user's private key. This ensures that all data updates are trackable, verifiable, and resistant to tampering. When a user creates or modifies content on a platform that operates within a decentralized ecosystem (such as Bluesky using AT Protocol), the content is initially saved to the user's PDR (hosted by their set PDS) instead of a centralized database (as is typical for traditional social media platforms). Since any PDR is tied to its user's DID, it allows the owner to migrate their identity and data to a different (compatible) PDS at any time, needing only to update the associated DID to the new location. Additionally, if the user backs up their PDR, they can use this backup variant in case of a PDS failure, accidental deletion, or any other similar issue by simply updating the DID document. This system offers resilience and portability, allowing users to maintain their social presence across platforms and services using AT Protocol.

Federation:

Once content is added to a user's PDR, federation mechanisms such as HTTP requests and WebSocket connections are used to replicate the updated data to other services within the decentralized ecosystem (in Bluesky's case, the atproto network). This replication/synchronization is based on what services exist within the network, relay services (indexers) such as Firehose continuously fetch updates from user PDSes, aggregating data based on what is required by the network's services. It removes the need for centralized control by allowing external applications and services to fetch, cache, and process user data independently.

One such primary service within atproto's network is App View, which is used to provide user-facing interfaces (Bluesky's native feed, blog clients like WhiteWind, Frontpage, Smoke Signal, etc.). It relies on specific categories of data such as records for: basic user identity, account state, posts, replies, engagement metrics, media blobs, interaction metadata, and moderation labels. The Bluesky App View will pull lexicon-based records under the *app.bsky.** namespace, such as: *app.bsky.actor.profile*, *app.bsky.feed.post*, and *app.bsky.feed.like*. In contrast, the WhiteWind App View will pull records under *com.whtwnd.blog.** namespace, including: *com.whtwnd.blog.entry*, *com.whtwnd.blog.defs*, and (query) *com.whtwnd.blog.getAuthorPosts*. Despite the two being the same type of service, they are customized for each platform; however, they may also pull non-app-specific records such as moderation labels (*com.atproto.label.defs*), which can be used for client-side filtering of certain topics, spam, hateful, or harmful content. Another example of a key service within the network is the Big Graph Service, it being responsible for indexing the network's global social graph and monitoring user relationships (follows, unfollows, mutes, blocks). Just like in App View's case, there can be multiple BGS instances across the network, each being tailored to construct a different view of the social graph, for example, the social graph of users within a specific platform, such as Bluesky.

Interoperation:

Another fundamental principle of the AT Protocol is interoperability, which refers to the ability for independent servers and applications to act upon user data in a consistent way. This is possible due to the implementation of a global schema network called Lexicon, which specifies the structure, naming, and transmission of different record types across the network, thus unifying the names and behaviors of the calls across servers. Each server or service implements a selection of lexicons, based on which determines the features they can support. For example, they can implement the core atproto lexicon to support user repository synchronization, or the Bsky lexicon to support basic social Bluesky behaviors. This architecture allows for a diverse range of applications (microblogging platforms – Bluesky, long-form blogs – WhiteWind, media-sharing apps – PinkSky, livestreaming services – GrayHaze, and more) to use the same core system for user identity and data. As elaborated previously, each platform/application can create and use its own custom Lexicon, just as Bluesky uses *app.bsky.** and WhiteWind uses *com.whtwnd.blog.**, each Lexicon being saved in the user's PDR, thus being accessible to any other application/service that supports the custom schema.

Instead of taking on a traditional web document-exchange style (HTML, XML, XHTML) approach, atproto exchanges schematic and semantic data. Which means that data is transmitted in a structured, machine-readable form (defined by the Lexicons), allowing vastly different applications/services to interpret each other's content while still performing on the same shared data record set. Thus, client applications can interpret and display content in their own unique ways, regardless of the server that hosts the data.

Achieving scale:

As mentioned previously, atproto facilitates a vast, distributed social ecosystem, which continuously scales horizontally. Such a large, ever-growing environment also comes with an ever-growing number of requests as well. For such a system to thrive even while perpetually expanding, it splits responsibilities among modular, interoperable services. Also mentioned previously is that at the core of this architecture are PDSes, which are in charge of managing the user's identity (including cryptographic record signings), hosting the user's PDR, and coordinating interactions with other services; but also allow users to configure their server's connections based on their preferred services. To achieve the monumental scalability necessary, the protocol introduces Relay services, which handle large-scale aggregation and discovery tasks that would otherwise overload servers.

Relays are optimized to process high-throughput events; they can be scoped to a platform, type of content, region, language, topic, or even an entire open network, essentially acting as middlemen between servers and services by indexing and caching global data from PDSes for services to request and use (App view, Feed Generator, Labler). Any user has the possibility to customize and run their own Relay, for example: Firehose Relays continuously listen for user PDR updates (such as posts, likes, follows) and streams the updates to subscribed services; Metric Relays track metric elements (likes, reposts, replies) to aggregate post/user interactions and provide metrics (profile statistics, trending feeds); Search-Optimized Relays index text from profiles and posts (user bios, content, hashtags, mentions), to allow fast user/hashtag/post searches; Feed Distribution Relays cache and sort posts (based on language, region, interest), resulting in more cost-efficient custom feed generation; Label Indexing Relays indexes incoming content and applies certain labels to assist with moderation and user-defined preferences; etc. Thus, the usage of both PDSes and Relays is what allows AT Protocol to scale in a decentralized manner, one managing user-specific operations and the other global-scale services.

Algorithmic Choice:

As briefly suggested in the previous section, the AT Protocol allows for the decoupling of content from its discovery mechanisms, treating content ranking and aggregation as modular services which users are free to select or replace (different App Views, Feed Generators, etc.), each having their own aggregation logic, and being maintained by an independent provider. By allowing users to customize their interactions within the network (through the configuration of preferred PDS-service request routes) without modifying the underlying content, this model eliminates any classic algorithmic social media concerns such as bias, manipulation, and filter bubbles. For example, Bluesky social media users have the option to use the platform's default feed ranking, or they could opt for a third-party Feed Generator, which prioritizes certain values/topics, or a generator that focuses on chronological order, displaying new content rather than trending content, and so on. Similarly, search and discovery tools can be swapped out, allowing for the filtration of specific topics such as academic content, language-specific trends, or perhaps even enabling a child-safe search variant. Due to these services operating on public, signed data from on-network PSDes, they can be built and hosted independently, not only allowing, but rather encouraging users to shape their online experience based on their personal views and specific needs within any of the applications using atproto.

Account portability:

As noted earlier, unlike standard social media platforms, instead of storing user identities and data within a singular, centralized, platform-owned database, the AT Protocol allows for the respective user elements to be hosted on user-defined servers. This provides users with full control over their identity and content at any time, allowing them to backup and move their PDRs across their chosen servers. Such a feature is possible mainly due to the usage of DIDs and DID documents, where the user has an assigned DID (a globally unique, cryptographically verifiable identifier) representing them across the network, and a linked DID document which hosts two public keys (a recovery key and a signing key) and the PDS address. The signing key is held by the PDS so that the server can validate changes to the user's data (PDR) and identity, whereas the recovery key is held by the user, allowing them to override the server's signing key, and thus providing a failsafe in case of PDS compromise or unresponsiveness (based on the predetermined security delay – typically 72h). And so, if the user backs up their PDR, then in case of sudden server issues, they can migrate to a new provider by uploading the backup PDR and reasserting their DID using the recovery key. The reassertion is done by creating a new signing key for their repository, providing it to the new server, and associating the new PDS address with their DID. This data and identity decoupling from platform infrastructure gives users control and allows them to preserve their social identity within the network, therefore also allowing for long-term reliability and trust.

Speech, reach, and moderation:

Another key principle of the AT Protocol is the deliberate separation of "speech" from "reach". This design choice is intentional, allowing both open participation and flexible, responsible content distribution. The speech layer is permissive, allowing anyone to post anything via their PDS (the content being stored in their PDR). The reach layer is selective; each client, service, or platform can decide on the content they want to display, prioritize, filter, or suppress, based on its own rules or moderation logic. This layered model system ensures that all users have the right to publish whatever content they may desire, but no content can be forcefully shown to other users. Likewise, users also have the right to view whatever content they may desire by configuring their services to display even restricted content. While speech and reach are decentralized layers, moderation is distributed and modular; hence, there's no single regulatory body. Each participant can be held liable for their part in the data's lifecycle – the publisher (user) for creating it, the PDS for hosting it, the service (App View, Feed Generator, etc. or Relay) for redistributing it, and finally the viewer for knowingly accessing it (a user going out of their way to configure/select services to view the restricted content). Ultimately, atproto allows users to post and view whatever material they want; however, it does not shield these users from possible consequences.

By integrating communication support for AT Protocol and Bluesky via the ATP API (*@atproto/api* package), SocialSentrix is able to create an new instance (*agent*) of the *BskyAgent* class, which provides access to a wide assortment of endpoints. Through these endpoints, the web application is now able to manage sessions and account authentication (*.resumeSession*, *.login*, *.session.did*), retrieve both public and private profile data (*.resolveHandle*, *.getAuthorFeed*, *.getFollows*, *.getFollowers*, *.getProfile*, *.getActorLikes*), and more (*.follow*, *.deleteFollow*, *.updateHandle*, *.upsertProfile*, *.getPostThread*, *.getRepostedBy*, *.getSuggestions*, etc.). These functionalities are essential for data collection, allowing Bluesky profiles to be integrated into frontend visualizations and backend SETIC scoring.

4.3. Network Dynamics and Semantic Structures

Social media spaces are far more than just account content assignment, metric stockpiling, and handle recognition; they also involve the analysis of how their users fit into larger network structures and how these structures affect connections, behavioral patterns, and overall reputation within the platform. The small-world phenomenon is a network science concept that describes how, even in large-scale networks, the majority of data nodes (in social media's case, users and content) can be reached through a small number of steps. Numerous existing social networks exhibit this vastly interconnected pattern, characterized by short average path lengths and high clustering coefficients. It also describes how information, behaviors, and influence quickly spread, often through a set of central or bridging users who, through their content and interactions, link communities that would be otherwise isolated from one another. Aside from this vastly structured web of nodes, ontologies such as FOAF offer a semantic framework, prompting machine-readable identity, relationship, and content modeling. Each FOAF profile can describe a person, their characteristics (metadata, bio, general metrics), and social connections (followers and following). Since a FOAF profile can be built on an RDF, it allows for relationships to be represented as linked data, forming semantic social graphs which can be extended across platforms and domains, allowing these networks to be queried and merged. Resulting in user identities that are not simply formed by direct ties between specific accounts, but rather the interconnected metadata and content characterizing a person's online presence.

Combining models such as these aforementioned ones could provide SocialSentrix with an even stronger foundation for deeper influence and credibility modeling. The SETIC components could make great use of not just raw metrics, but also data based on analyzing a user's position within a generated social graph and their semantic links to other users and across different platforms.

4.4. Modular Interface Design with Micro-Frontends

As web applications become more and more complex with each update, micro-frontend architecture has become a reliable and scalable method for organizing the frontend flow and codebase. In a similar manner to SocialSentrix's component splits and modularity, the micro-frontend model proposes that the interface and flow of an application be divided into components that are then developed as independent mini-applications, which can then be combined to form a fluid UI. If these components function independently, it allows for much faster development and bug fixing, as the work can be split and parallelized, resulting in far faster update rollouts, issue handling, and bug patches. Taking SocialSentrix's modularity further to a micro-frontend approach would allow for even easier platform integration and development, ensuring the platform's maintainability, scalability, and adaptability even after numerous updates and feature additions.

5. System Design and Implementation

SocialSentrix was developed using a modular client-server architecture, purposefully designed to guarantee cross-platform compatibility, extensibility, and maintainability. This architecture consists of three main layers: a React-based frontend, a Node/Express JS backend, and a MongoDB document store. This multi-layer separation facilitates code reusage, efficient debugging, and feature evolution without having architectural fragility as a cause for concern. The frontend functions as a responsive, single-page application, hosting all of the user interaction and data visualization. While the backend serves as a platform-neutral logic core, offering RESTful endpoints that connect to APIs and scraping tools to extract data for a given profile from social media platforms such as Reddit and Bluesky. The MongoDB layer stores raw and processed data in a primary database, consisting of four distinct collections. These collections host various types and quantities of data, including user accounts and sessions, profile metadata, extracted profile content, derived metrics, and more. This chapter presents a structured walkthrough of SocialSentrix's system design, with each subchapter describing a major subsystem or design choice.

5.1. System Architecture and General Application Flow

This subchapter goes into greater detail regarding SocialSentrix's runtime behavior, with a particular emphasis on the data flow, request routing, and component organization that support the user interface and backend processing pipeline. At its core, the web platform uses an event-driven, modular flow, which begins with the user's connection and ends with interactive, data-rich visuals. The process can be visually seen in [Figure 5.] & [Figure 6.], and broken down into the following stages:

5.1.1. Frontend Input Capture

Once the user accesses the web application, they are automatically directed to the homepage ("/" route), where a dynamic input form is rendered by the included *SubmitProfile.jsx* component. This component supports multi-profile submission and the login mechanisms for multiple platforms (currently Reddit and Bluesky). It also offers account-based real-time profile input completion, where if a user is logged into their SocialSentrix, they can directly input a profile (via username, handle, or link) or select one from their list of previously submitted social media profiles.

5.1.2. Request Construction and Dispatch

Continuing from the *SubmitProfile.jsx* component, once the profiles are submitted, the input string is then split (using the '&' character), where these splits are then set into a *profiles* map. Based on the resulting *profiles* map, it constructs concurrent POST requests to the */submit-profile* backend route using *Promise.all*(). Each request includes the platform type (as '*platform*'), the profile input (as '*input*'), alongside the user's cookies (via "include: credentials").

5.1.3. Backend Routing and Dispatch

The submitProfile.js route handler begins by extracting the clean username from the *input* via platform-based regex validation. Afterwards, it verifies the existence of a social media profile based on the *platform* and extracted username (via an API / Puppeteer check). If this platform-based check fails, the system then verifies whether the profile has been previously stored in MongoDB (under *Users/profiles*). If this check also proves false, then a fallback is triggered, returning an error message; otherwise, the process continues as normal. The request is routed to the appropriate controller (*RedditProfile.js* or *BlueskyProfile.js*) based on the platform parameter. Each controller includes either or both platform-specific API calls and Puppeteer mechanisms, allowing authentication simulation (through the usage of the user's saved account tokens) and ensuring thorough profile data retrieval (API call-based data returns, crawling auxiliary public content, and profile metadata). Afterwards, all the collected data is then parsed using the *buildMeaningfulUpdate()* function, which filters out empty fields before saving or updating the profile's structured document (JSON). This saving/updating is realized in MongoDB, within the *Users* database, under the */profiles* collection. Finally, based on whether the social media profile is owned by the SocialSentrix user, the corresponding *guestView* and *ownerView* data structures are returned as the response.

5.1.4. Frontend Rendering and Interaction

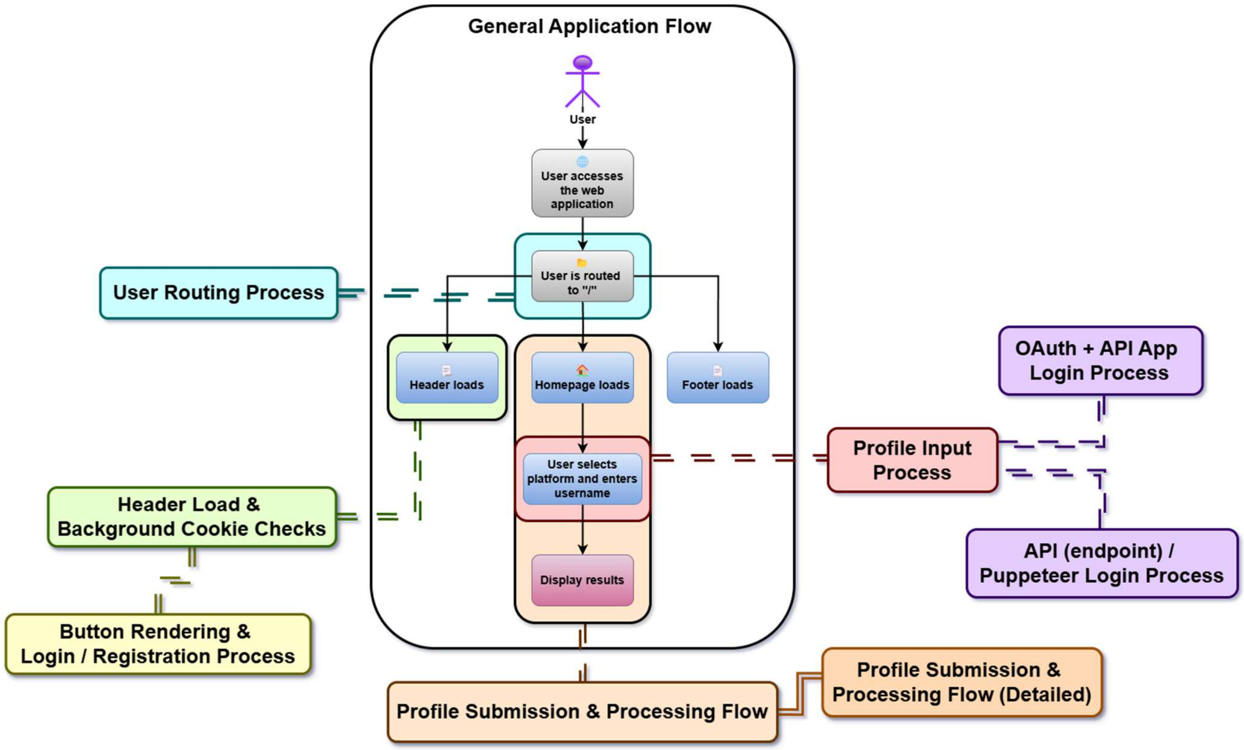
Once the response is received from the backend, the frontend launches an instance of the *ResultedWindow.jsx* component, rendering it alongside *SubmitProfile.jsx* within the main *Home.jsx* layout. For each response (profile) received, a new *SortableResultWindow.jsx* subcomponent is added to the rendered *ResultedWindow*, effectively generating a new window for each given profile. Additionally, within SortableResultWindow, multiple sub-windows are dynamically created using the *SubWindow.jsx* and *ChartBlock.jsx* subcomponents. With corresponding charts and user interface elements, these sub-windows are customized for specific content categories, including posts, comments / replies, upvotes, downvotes, and reposts. Alongside these graph-populated sub-sections, a final sub-window is added, where, based on the selected dates, at the click of a button, the SETIC reputation score calculation commences for the given profile. This process begins by sending a *fetch()* request to the platform-specific controller endpoint (*/<platform>/setic*).

5.1.5. Data Transformation and SETIC Scoring

Once the request is intercepted by the controller (*.../setic*), the backend begins by retrieving the profile's *guestView* data section from MongoDB. However, in Reddit's case, before the SETIC calculation function is reached, an estimated execution time is calculated based on factors such as the content volume (estimated batches), account token validity, number of unique subreddits, and expected rate limits. Once the *optimisticETA* and *pessimisticETA* are determined, they are returned to the frontend, where the user is prompted with *alert()*, containing the summery of the profile's account token validity (valid, expired, or missing) and the estimated computational time interval. Upon the user's confirmation to proceed, the backend continues with the calculateSETIC(). Initially, within this function, the raw profile data is filtered by content type and the selected time range. The resulting dataset(s), including filteredPosts, filteredComments, etc., as well as the complete profile dataset and account token, are then passed to individual SETIC functions, responsible for computing the Sentiment, Engagement, Trustworthiness, Influence, and Consistency scores. Finally, once each function completes its computation, the resulting values are then parsed (along with some specific insights) and aggregated to determine the profile's overall reputation score. Afterwards, the whole collection of data is returned to the frontend, where the user can now view the total reputation score, each individual SETIC score, and additional insights, providing the analysis with transparency and interpretability.

Figure 5. End-to-End Application Flow

Figure 6. Minimal Overview of SocialSentrix's System Architecture



5.2. Tools and Technologies

The SocialSentrix platform is developed using an entire ecosystem of chosen technologies, frameworks, and tools that collectively support the goals of cross-platform data collection, reputation modeling, interactive visualization, and secure user interaction. This section provides an in-depth overview of the software stack and development tools used in the frontend and backend components of the project. Each technological incorporation is based on the tool's functionality, scalability, developer productivity, and applicability relevant to the specific methodological requirements of reputation analytics.

5.2.1. Development Tools

The development of SocialSentrix was supported by a set of modern tools designed to enhance efficiency, code quality, and project management. The primary development environment was Visual Studio Code (VSCode), which was expanded with a plethora of plugins to improve functionality. Some of the more notable extensions are GitHub & GitLens (for commit and code history), AppMap (for backend flow tracing), ESLint (for code formatting and linting), DotENV (for environment variable management), IntelliSense & Path IntelliSense (for code and path completion), Tailwind CSS IntelliSense (for CSS utility). Additionally, the development process was enhanced by a number of third-party applications and frameworks, such as Adobe Photoshop for asset design, GitHub Desktop for streamlining Git operations, Draw.io and Mermaid for diagram creation, and Trello for project planning, task management, and progress visualization. These tools simplified tasks that, while non-code, are essential to the overall development process. Together, all of the mentioned tools, plugins, and others provided a strong basis for the effective development and project management of SocialSentrix.

5.2.2. Web Frameworks and Application Structure

The application is built on a component-based architecture, divided into logical, reusable units. The frontend of SocialSentrix was built using React, a declarative JavaScript user interface library. Some of the main components are the header (global), footer (global), and home (route), where the submit profile & result window sub-sections have their own subcomponents, such as profile parsing, graph rendering, login/registration popups, and others. Some of the key libraries included are React Router DOM (for client-side navigation), React-Icons (for iconography), Tailwind CSS (for styling and responsive design), Recharts (for data visualization & charting), and many more. The backend is built using Node.JS with Expres.JS as the web framework, which handles all of the routing and API logic. A few of the essential packages include Axios (for HTTP requests), CORS (for cross-origin resource sharing), MongoDB (for database operations), Puppeteer (for web scraping and automation), and Vader Sentiment (for sentiment analysis processing). While also having a modular directory structure, with separate route files for each platform (Reddit, Bluesky, Twitter), and matching controller files for their authentication method, profile crawling, and SETIC score calculation. This separation of responsibilities supports easy scalability and debugging.

5.2.3. Data Storage and Retrieval Technologies

SocialSentrix uses MongoDB, a NoSQL document database used for persistent data storage. Since traditional relational databases (such as MySQL, PostgreSQL, etc.) would need highly intricate schema modifications for different data formats across platforms, MongoDB was chosen as the main data storage due to it easily being able to store heterogeneous and nested data structures typical of social media profiles and activity logs. Its flexible document structure allows for far more efficient (and mass) data storage, update, and retrieval, without the need for constrictive and/or complex normalization. The backend interacts with MongoDB using the official MongoDB driver, where mongo.js is used to implement custom database wrappers that simplify operations and offer reliable error handling (connectMongo, getDb, dbAccounts, etc.). The necessary data, such as user-related data, user profiles (social media profiles), registration and login requests, are all categorized and stored under the same database ("Users"), but under different collections ("/profiles", "/accounts", "/requests", "/pending"). For example, user-related data, including registration information, associated profiles, and admin sessions, is stored in its dedicated collection Users/accounts. In order to differentiate between authenticated and public scraping results, the system implements a dual-view model (guestView, ownerView) for all given profiles. Additionally, the backend implements a retrieve-update-result system in which, when a user sends a social media profile, instead of the data being simply returned, we update the content for the given user profile in /profiles, and then send the guestView and/or ownerView profile data. This method ensures up-to-date profile data and a fallback in case there's an unforeseen issue with the social media profile or platform.

5.2.4. Data Collection and Privacy Practices

Data collection is initiated when a user submits a social media profile for analysis (via the SocialSentrix platform or directly through a HTTP request). When available, the backend uses public API endpoints (Reddit's JSON API, Bluesky's ATProto API, etc.), if the platform has no public API endpoint, or we require a deeper scraping (for example: Reddit's achievements, moderator roles, etc.), then a headless browser automation through Puppeteer is performed. Logging into a social media platform is completely optional and at the user's discretion. If they do indeed log in, no data is saved aside from necessary authentication tokens, which are then assigned to their SocialSentrix account. For extra security, these tokens can be encrypted and decrypted before saving / usage, where, for our current demo, while not in use, this extra layer of security has been added. The platform handles mostly user-visible or publicly marked data, and clearly separates any extra metrics / data extracted from the user's platform-logged in account. Furthermore, SocialSentrix closely monitors user sessions and accounts, having constant checks on the validity of their sessionID and userID cookies, so that all data access and requests are securely controlled. Additionally, by also checking account permissions whenever a request is made (directly in the backend), we prevent any unauthorized requests.

5.2.5. Data Cleaning, Normalization, and Structuring

To ensure consistency, once the data is collected from the respective platform, it undergoes a cleaning and normalization process. During this process, redundant fields are removed, dates are parsed into ISO strings, and nested structures (nested comments, reposts, etc.) are flattened for analysis. To prevent the database from becoming bloated with duplicate elements or suddenly having missing entries due to platform or communication errors, the received data is parsed through the buildMeaningfulUpdate() function, where it compares the new data with existing data in MongoDB, thus updating only the modified (non-empty) fields. In order for the SETIC scoring framework to execute time-bound computations, the profiles' activity is converted into time-series data objects, categorized by day, week, month, etc. This structuring is crucial for providing meaningful analysis and supporting the frontend's visualization filters.

5.2.6. Sentiment Analysis and Natural Language Processing

Within the SETIC framework, the sentiment score is determined using sentiment analysis via VADER (Valence Aware Dictionary and sEntiment Reasoner), a lexicon and rule-based sentiment analysis tool tailored for social media text. This library is specifically attuned for social media texts due to its vast case handling of contexts and sentences frequently present within such social environments, such as: the translation of emojis, sentiment-laden slang, specific slang word modifiers ("kinda", "meh", etc.), degree modifiers (for sentiment intensity), sentiment-laden initialisms and acronyms, punction-based sentiment intensity modifiers ("???", "!!!", etc.), word-shape based emphasis, typical negations, and more. More sophisticated models, such as BERT or domain-specific transformers, might be incorporated in later versions to improve multilingual support, sarcasm detection, or contextual understanding. For the time being, VADER offers a solid basis for real-time reputation estimation due to its quick performance, impressive and reliable accuracy.

5.2.7. Visualization and Frontend Interaction Libraries

To render visual insights, SocialSentrix employs “Recharts”, a React-based composable charting library that generates responsive line and bar charts based on the resulting data after profile submission. Through interactive controls, users can group the data by time units (minute, hour, day, week, month, year), switch between metric representations (number of posts / comments / likes / etc.), select custom datetime ranges, and interact with graphs through zooming and panning functionalities. Throughout the platform, “Shadcn/ui” components are incorporated into the UI to maintain consistent form controls and design elements. Drag-and-drop capabilities provided by “dnd-kit” improve the platform’s interactive functionality even further, by allowing users to rearrange dashboard elements (chart / result sections) and personalize their analysis view. Additional features include customized tooltips and point-level data overlays, responsive styling through Tailwind CSS, and much more. This combination of visualization tools and interactive elements enables users to perform comprehensive profile analysis with intuitive navigation and clear data presentation.

5.3. Session Validation, Management and Email-Based Authentication Flow

Unlike traditional password-based authentication systems, SocialSentrix implements a simplified, email-based registration and login process designed for simplicity, security, and ease of use in collaborative settings. This system not only enhances user privacy and prevents potential credential theft (through the use of token-based one-time email confirmations), but it also facilitates easier access for corporate teams and research groups to a single, centralized account for multiple users and sessions. There are three main components to this part of the system: header-based cookie detection and UI state handling, frontend registration and login popups, and backend account management and session confirmation, which are handled across *Header.jsx*, *PopupLogin.jsx*, *PopupRegister.jsx*, *PopupUser.jsx*, *accountRegisterConfirm.js*, *getUserData.js*, and *loginStatus.js*.

5.3.1. Header Session Detection and Access Control

Once the web application loads, the *Header.jsx* component begins with two parallel processes in order to determine and display the user’s authentication status. The first *useEffect* hook calls *validateSessionCookies()*, which sends a backend request to */mongodb/validate-*

cookies, in order to confirm the legitimacy of the *userID* and *sessionID* cookies (if they are set). Based on the response, the frontend updates its state and sets the user information for later use in the UI. The second *useEffect* hook checks if the user has a *sessionID* cookie set in their browser, to conditionally launch a polling mechanism via the *startLoginStatusPolling()* function. This polling mechanism queries the backend every fifteen minutes using the */mongodb/check-login-status* endpoint to monitor the status of the user's account login request (within MongoDB, under Users/requests). Depending on the returned status, the system may take one of the following actions: if the status is "Pending", it continues polling; if the status is "Expired" or "Denied", it clears the *sessionID* cookie and alerts the user to send a new login request; if the status is "Confirmed", it sets a new *userID* cookie, notifies the user that their login request was accepted, and asks whether they would like for the page to be refreshed (so that certain states and UI elements may be updated).

Furthermore, this logic is directly tied to the visual state of the header, where, depending on the login state of the user, it conditionally renders the icon (question mark icon / user icon) and sets the PopupPortal. Based on the set user state, on click it launches *RegistrationPopup.jsx* through the set React Portal, hosting either *PopupRegister*.jsx or *PopupUser.jsx*. This layered mechanism allows SocialSentrix to maintain a stateless yet persistent authentication flow, where sessions are dynamically verified on demand, as well as allowing effortless yet reactive transitions between registration, login, and user data visualization.

5.3.2. Registration and Login Submission Logic

Once the user initiates an authentication action by clicking the login / register button (which appears by default when a valid *userID* and *sessionID* cookie pair is not present), the *RegistrationPopup.jsx* component is rendered via a React Portal in *PopupRegister* mode. In this registration view, the user is prompted to provide a valid email address for account creation. Both the email and confirmation fields are validated and regex-checked in real-time, whereupon submission, a POST request is sent to the */mongodb/register-account* endpoint. Here, the backend checks whether an account with the submitted email already exists under MongoDB's */accounts* collection. If so, then the user is prompted by the frontend to log in instead. Otherwise, a new document is added to the */pending* collection, containing the account's generated UUIDv4 confirmation token, email address, and registration request timestamp. An email is then composed using nodemailer, containing a brief message body, the generated token-based *confirmationUrl,* and *expiresAtString* datetime, which is then sent to the provided address.

Additionally, while the *RegistrationPopup* is rendered, the user can toggle between the component's *PopupRegister* and *PopupLogin* mode. Similarly, as it can be seen in [Figure 7.], in the login view, the user is prompted to enter their account's email address, name, and optionally, a comment to be added to the login request. As with the registration form, the email is regex-checked in real-time, alongside the name field, which is checked to be non-empty. On submission, a POST request is sent to the */mongodb/request-login* endpoint, where the backend first checks in MongoDB's */accounts* collection for the existence of an account with the given email. If no account is found, then the user is prompted by the frontend to register instead. Otherwise, the login process continues as expected, where we generate and set the data for the request under the */requests* collection. The newly added document contains a UUIDv4 *sessionID* (which is also set as a cookie), the given email address, name, and comment, as well as the requester's user agent, IP, location, a "Pending" status, request creation and request expiration datetime. An email is then composed and sent to the provided email address through nodemailer, as seen in [Figure 8.], it contains the message body, generated session-based *confirmUrl* & *denyUrl*, alongside the name, comment, IP, location, user agent, and request expiration datetime mentioned previously.

This authentication model prioritizes privacy and simplicity by allowing users to confirm their registration and login requests with a single click of a button within their inbox. This approach minimizes common vulnerabilities such as frequent password recovery, leaked credentials, brute-force attacks, and phishing-based theft. Furthermore, since the requests and credentials are tied to persistent cookies (sessionID), multiple users can operate in parallel under a shared account, enabling a more efficient system for associating and indexing social media profiles.

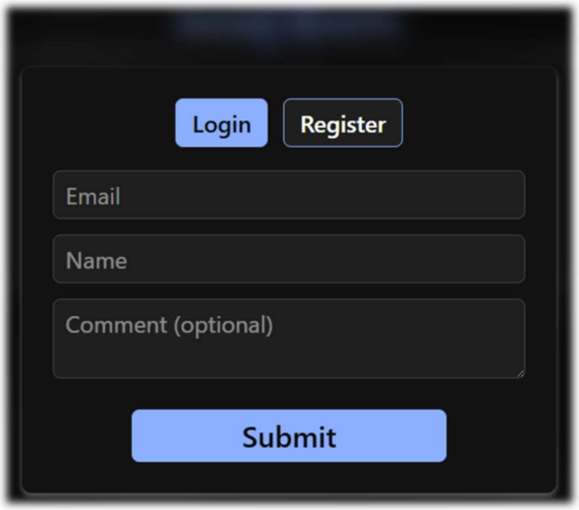


Figure 7. Web Application – Login Popup

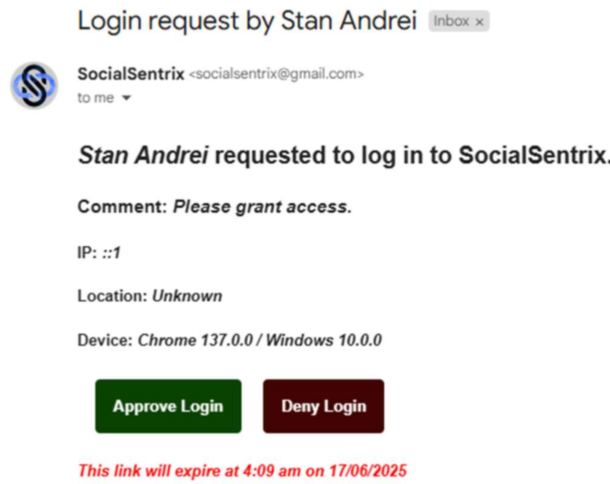


Figure 8. Web Application – Login Request Email

5.3.3. Email Confirmation and Session Issuance

Once the user receives the (register/login) confirmation email, they must click the embedded button to complete the account creation/authentication process. This button contains a custom token/session-based URL that directs the user to a backend endpoint (*/mongodb/confirm-account?token=<token>*, */mongodb/confirm-login?sessionID= <sessionID>*, /mongodb/deny-login?sessionID=<sessionID>) which processes the request and then redirects the user to the frontend */parse* route, displaying a visual response (confirmation, alert, or error) for the account confirmation / request acceptance or denial based on the backend response. For the account confirmation, the backend mechanism validates the provided token, where if it is valid, it removes the account from */pending* collection. It then creates a new document under */accounts*, containing a newly generated UUIDv4 ID, the email address, creation timestamp, and three empty arrays for *adminSessions*, *associatedProfiles*, and *ownedProfiles*. For the login confirmation or denial, based on the provided session ID, the backend mechanism checks if the request under */requests* has not already expired, been denied, or confirmed. If it is still valid, the request status is then updated to "Confirmed" / "Denied", and as mentioned before, the user is redirected to the */parse* route. As seen in [Figure 9.], it displays visual confirmation and then redirects to the homepage ("/" route) where the new *userID* cookie is set. This token/session-based confirmation system eliminates the need for credential storage while supporting possible future extensions such as device tracking and admin session flags. Furthermore, this modular coupling of the backend and frontend guarantees a smooth and responsive transition for all scenarios, fluidly guiding and informing the user.

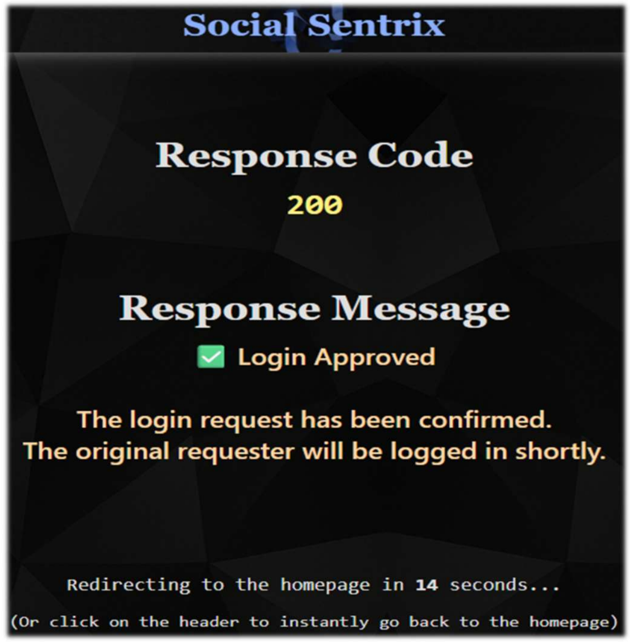


Figure 9. Web Application – Login Request Confirmation (/parse)

5.3.4. Session Management and Visualization

After the user successfully logs in, after a quick *userID* and *sessionID* pair identification and validation, they gain access to the */managesessions* route (*Sessions.jsx*), which is directly accessible by clicking the "Manage Sessions" button within the *PopupUser.jsx* interface (it now replaces the default *PopupRegistration.jsx* element). Using their set *sessionID*, the frontend communicates with the backend through the */mongodb/check-session-permission* endpoint, checking if the current user session has admin privileges. This verification is done by locating the *sessionID* within the *adminSessions* array of the given *userID* account. If it cannot be found, the user is notified via an alert that they do not have permission to visualize or modify the current accounts sessions. Afterwards, the alert asks the user if they would like to submit an admin permission request. Upon confirmation of the alert, an email is automatically generated and sent to the account holder's inbox. If the account holder chooses to grant admin privileges, a backend request is then created to the */grant-session-admin* endpoint. Here, the backend adds the provided *sessionID*, to the account's *adminSessions* array, thus providing the user session full access to the account's sessions. Now, upon accessing the */managesessions* route, after a quick permission check, a backend GET request is made to the */get-sessions* endpoint, returning the list of sessions and requests tied to the account (via email). Upon the frontend receiving the map of sessions, their status and data, are rendered visually by *Sessions.jsx*, providing the user with an interface, where they can manage said sessions.

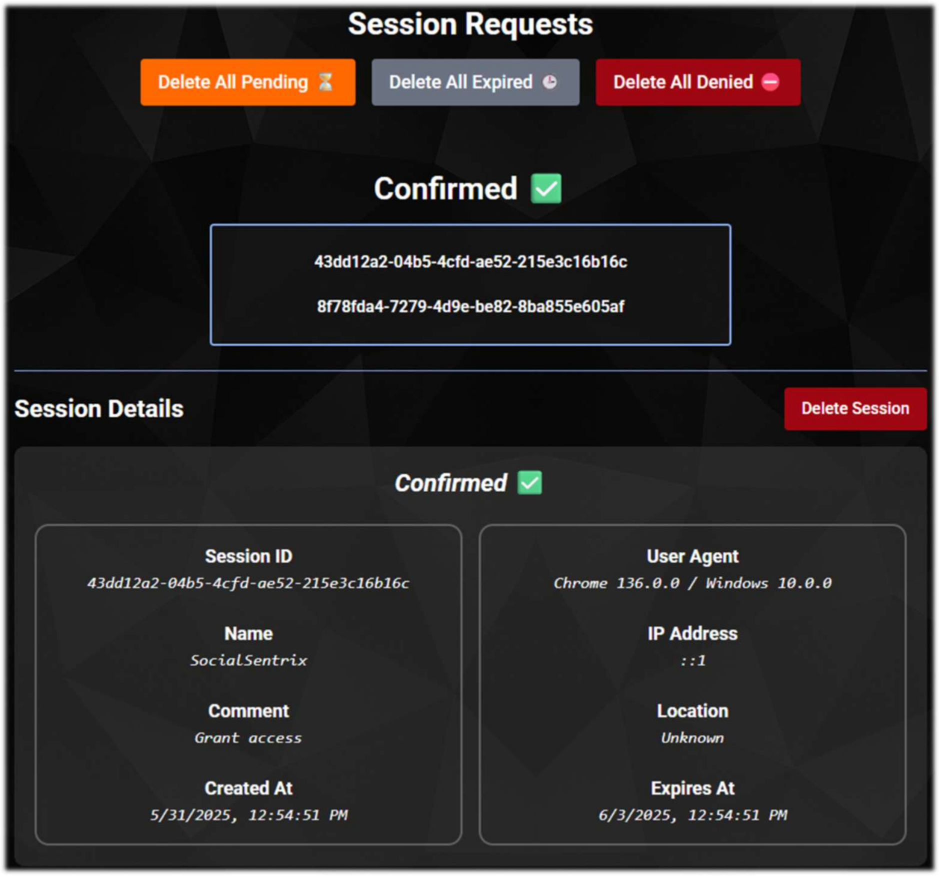


Figure 10. Web Application – Session Management UI

[Figure 11.] details the step-by-step process of an account registration/login, as well as the interactions between the frontend components (*Header.jsx*, *RegistrationPopup.jsx*, *PopupLogin.jsx*, *PopupRegister*.jsx), backend endpoints (Express), and included services (Nodemailer, MongoDB). It outlines how session-linked tokens are generated, confirmed via email, and then converted into authentication cookies, thus establishing a secure user session.

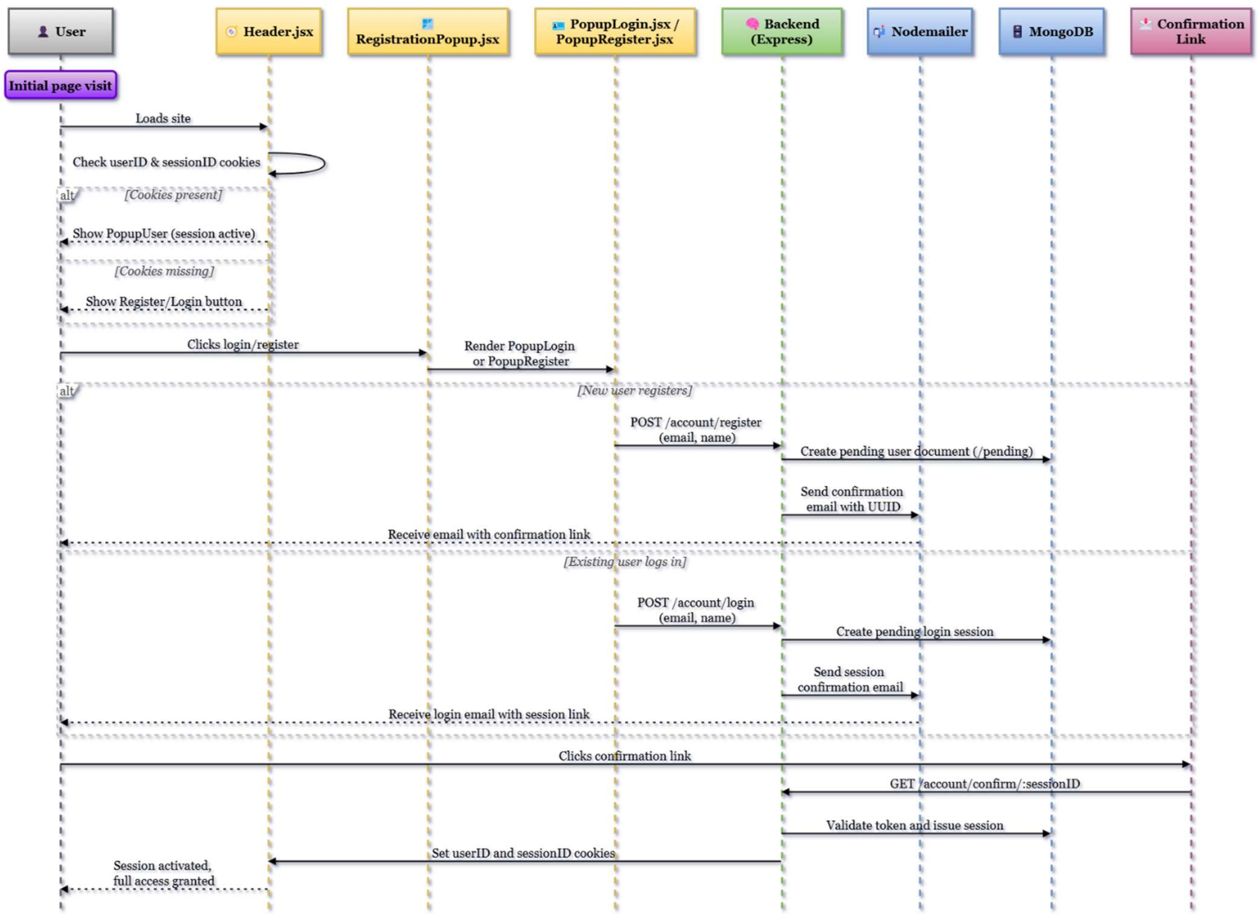
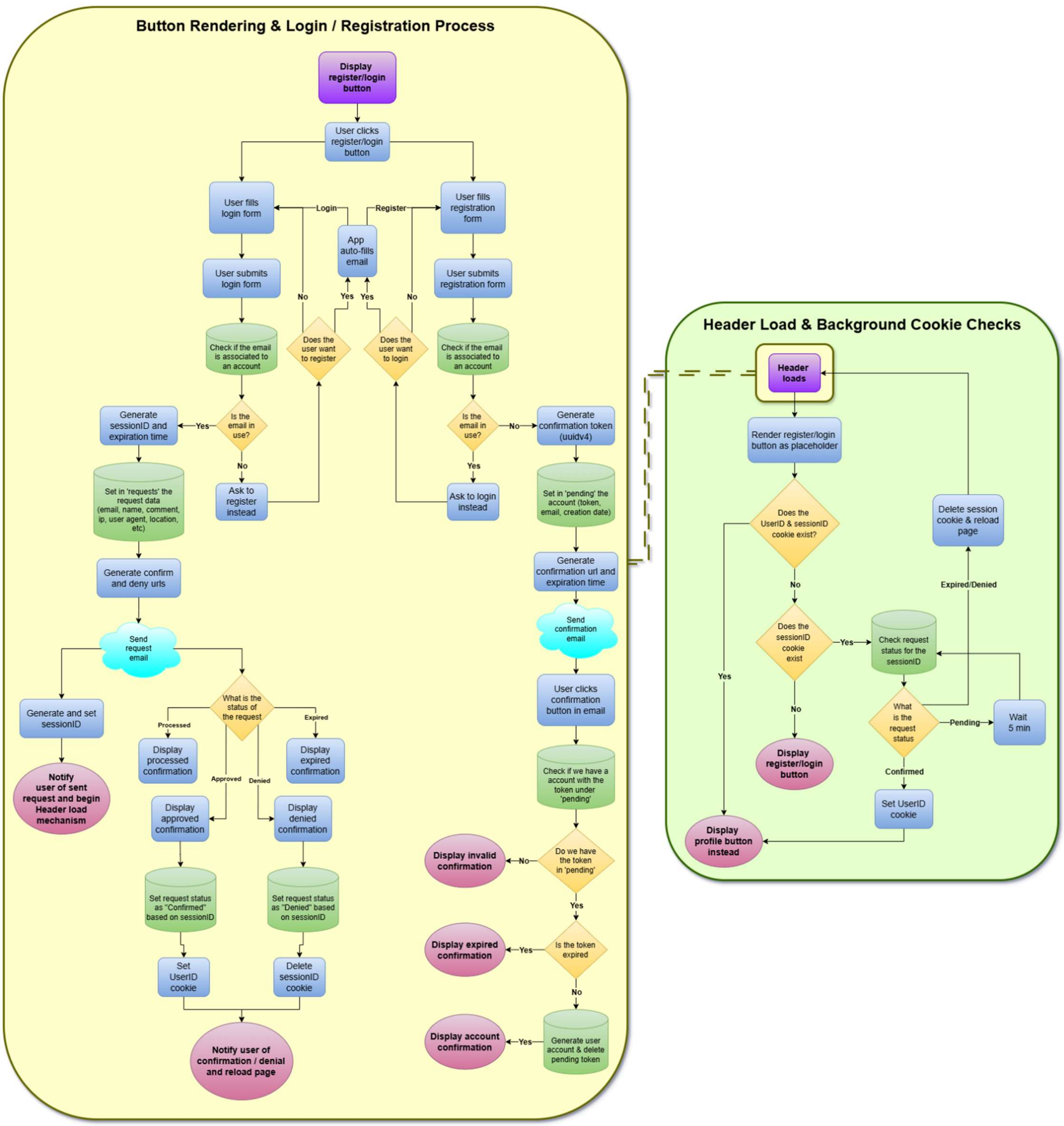


Figure 11. Authentication Flow

The flowchart presented in [Figure 12.] illustrates the global (header) background cookie validation, as well as the decision-making process underlying SocialSentrix's email-based login and registration interface. Guiding users through account confirmation, session validation, and possible fallbacks depending on session states, email existence, token status, and more.

Figure 12. Header Checks & Authentication Conditional Logic



5.4. Frontend Implementation and Interaction Design

Built with React, the SocialSentrix frontend prioritizes real-time visual analysis, multi-profile support, and modular interaction. It also includes dynamic login systems for each of its given platforms, whether it be through an OAuth process, API session instance, or simulated Puppeteer instance, thus allowing for token-based data access, profile ownership tracking, and authenticated data scraping. This section describes the frontend's handling of platform authentication flows, window behavior, user input, and result rendering.

5.4.1. Profile Submission and UI Behavior

On the homepage ("/" route), the *SubmiProfile.jsx* component provides users with a platform selection dropdown, where, after selection, as seen in [Figure 13.], it reveals an input field and platform authentication button/form. The input field allows for the introduction of one or more platform profiles (separated by "&"), where if the user is authenticated into their SocialSentrix platform and has previously submitted profiles or authenticated into a social media account, the component reveals a real-time suggestions list, retrieved from the account's *ownedProfiles* and *associatedProfiles* arrays. After submission, the input is parsed into a profile map, and a Promise.all() is used to make concurrent POST requests to the */submit-profile* endpoint for each profile within the map. Every request contains the selected platform, raw profile input, and attached browser cookies. If the profile's previously set token credentials are invalid, then it prompts the user to log in to the respective social media account once more.

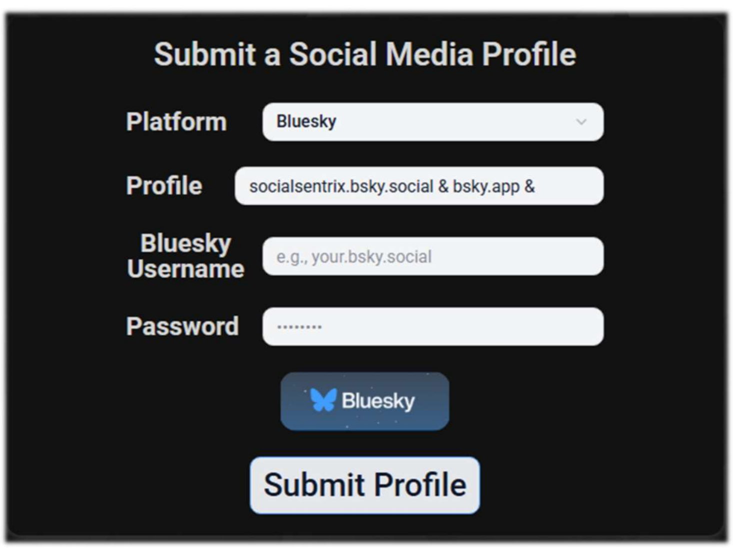


Figure 13. Web Application - Profile Submission and Platform Authentication

5.4.2. Platform Login and Token Capture

When the user initiates platform login – by clicking the platform-specific auth button, the frontend makes a request to the respective /<platform>/auth endpoint. Each platform uses a specialized authentication flow based on its API existence and capabilities. For instance, in Reddit's case (RedditAuth.js), it is done through the usage of a custom Reddit developer application, which allows for OAuth2-based login by redirecting users to Reddit's authorization URL with the necessary scopes (identity, history, etc). Once the user grants the developer application permission, the request is then returned to SocialSentrix's backend along with an authorization token, while the user is redirected back to the homepage. In the backend, the returned reddit_token, alongside the platform and social media username, are then stored as an object under the user's ownedProfiles array within MongoDB.

In Bluesky's case, we make use of the @atproto/api library to request session tokens (accessJwt and refreshJwt) and the account's did (decentralized identifier) based on the provided account login credentials (as seen in [Figure 13.]). However, instead of redirecting the user to and from the web app, it simply notifies them through an alert if the platform profile authentication was successful or not. Similarly to Reddit, in the backend, the token (accessJwt), refresh (refreshJwt), and did, are set as an object under the user's ownedProfiles array. Unlike the others, in Twitter's (X's) case, the platform login process is performed through an automated headless browser authentication using Puppeteer. After profile credential submission, the backend launches a headless Chromium instance, navigating to X's login page, and simulating the login process by entering the provided credentials. Once the profile login was successful, the auth_token and ct0 cookies are captured, alongside the determined profile handle, and with the platform, are set as an object under the user's ownedProfiles array. As in Bluesky's case, the user is not redirected anywhere, the entire process is automatic, and only alerts the user of the profile's login success or failure. Thus, this multi-modal authentication method reflects SocialSentrix's modular and extensible nature while preserving a standardized backend mechanism.

5.4.3. Dynamic Visualization, Graph Controls, and UI Behavior

Upon receiving an initial successful backend response, the main application layout (specified in *Home.jsx*) dynamically renders a new instance of the *ResultedWindow.jsx* component – it being the container responsible for displaying all of the submitted user profiles data. Each user profile is encapsulated within its own *SortableResultWindow.jsx* element, allowing for the profile-specific data to be further categorized into sub-windows (through multiple new instances of *SubWindow.jsx*), based on the content type (posts, comments, upvotes, downvotes, and reposts). Where every *SubWindow* contains a header with the category name and action controls (download, *compareMode* toggle, minimize, maximize, close) and a *ChartBlock.jsx* sub-component (which generates the visual graphics). The visual *ChartBlock* elements, seen also in [Figure 14.], are dynamically generated based on the provided profile's data, allowing each graph to be highly interactive.

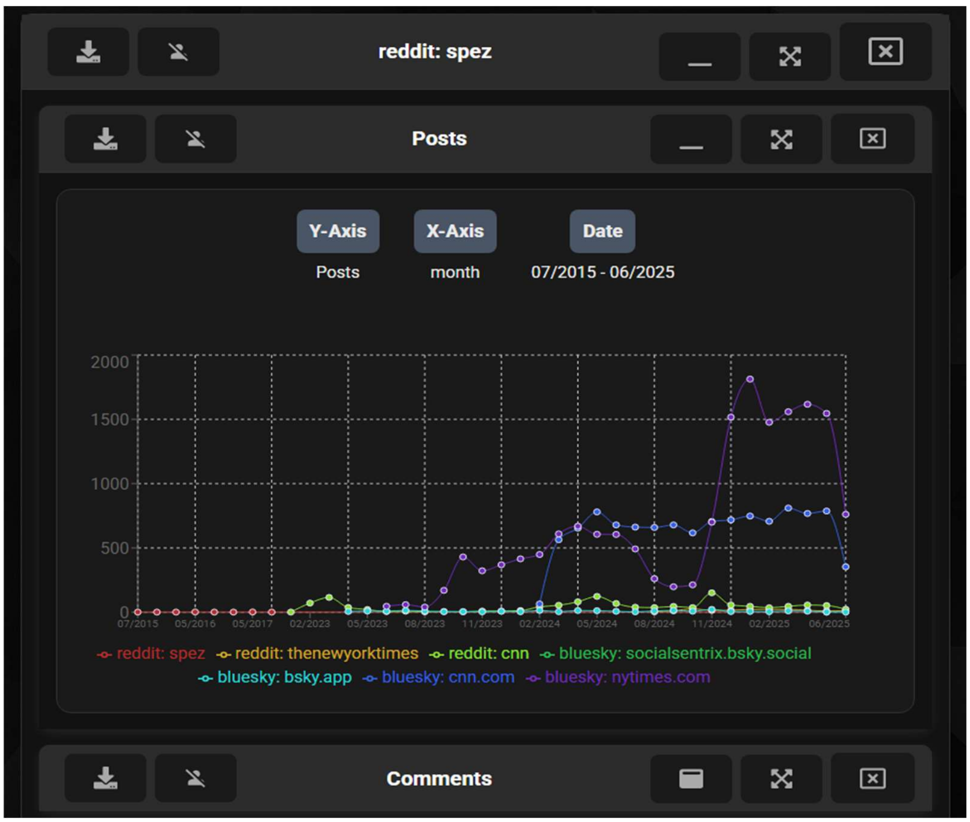


Figure 14. Web Application – Frontend Window, Sub-windows, and Chart Blocks

Some of the interactive capabilities and tools provided for these charts are:

- On focus – zoom in/out (mouse wheel based), as well as pinch gestures for the X-axis, allowing for the focus to be horizontally moved within the zoomed-in graph, making it possible to navigate through dense time-based datasets;
- Modifiable X axis, changed by selecting the desired granularity (minute, hour, day, week, month, or year) – as seen in [Figure 17.];

- Modifiable Y-axis, changed by selecting the desired metric (for posts: number of posts/comments/upvotes/reposts, for comments: number of comments/upvotes, etc.) – as seen in [Figure 15.];

- Selectable data time-range (by default, the oldest and newest content among the profile(s)), changed by selecting the *start* and *end* time range – as seen in [Figure 16.];

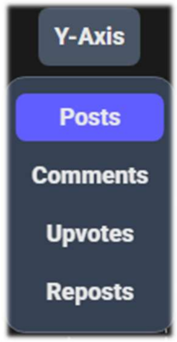


Figure 15. Web Application – ChartBlock Y-Axis Dropdown

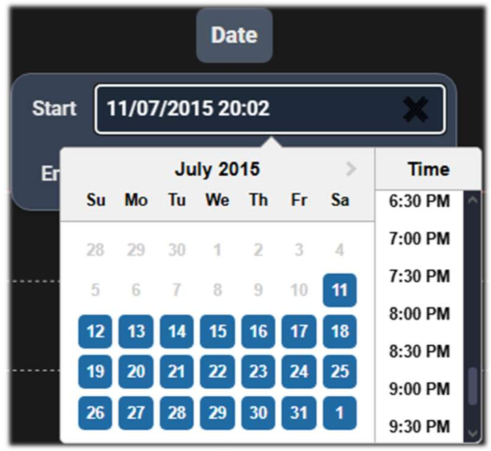


Figure 17. Web Application – ChartBlock X-Axis Dropdown

Figure 16. Web Application – ChartBlock DateTime Selection

- Dynamic tool tip which appears when hovering over any data point, displaying a simple UI element with the profile(s) and their metric numbers for the point (set based on the Y-Axis selected metric and sub-window category) – as seen in [Figure 18.];

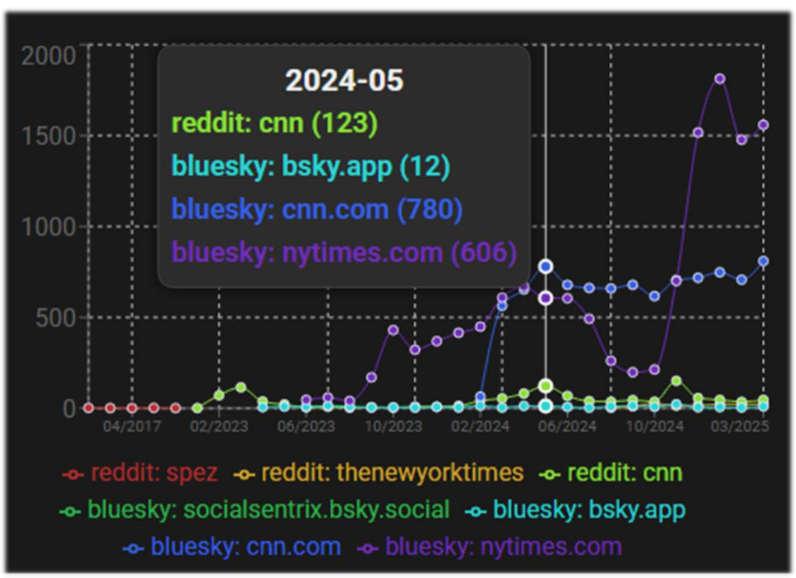


Figure 18. Web Application - Frontend Chart Tooltip

- On-click data visualization section for the selected data point, displaying the data for the given profile(s), underneath the graph;

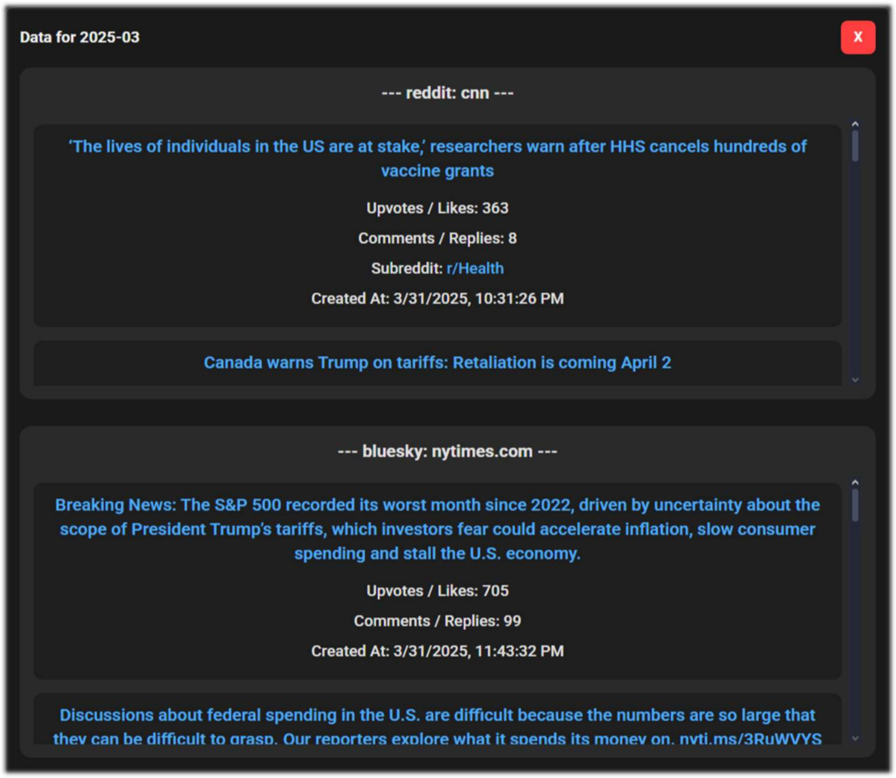


Figure 19. Web Application - Frontend Chart Data Visualization

- Chart download possibility, each chat can be downloaded in one of three formats and two modes:
  - PNG / SVG Selection - saves the current focused area of the chart as a PNG;
  - PNG / SVG Full - extends the chart width rendering based on the number of X-Axis segmentations (so that a distance is created between the data points), allowing the chart to be readable without having constant overlapping points when saving as a PNG;
  - CSV – saves the profile(s) datapoints and metrics, based on the given *X-Axis* granularity, *Y-Axis* metric, and selected datetime, as a raw timestamped dataset in tabular format;

For each *SortableResultWindow*, after the generation of the chart sub-windows, a final sub-window will be rendered, which, as seen in [Figure 20.], has the purpose of triggering and hosting the results of the SETIC score calculation (based once again, on the selected time frame). Once the scoring is triggered, the result – the profile's reputation score is then rendered under the *Calculate SETIC* button. This element, while initially unremarkable, has the ability to display on-toggle the score result of each SETIC component – as seen in [Figure 21.].

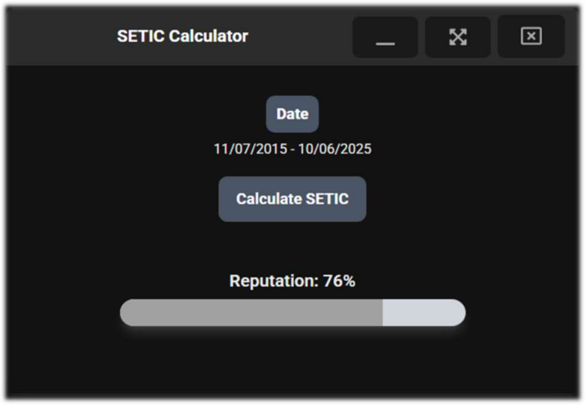


Figure 20. Web Application - SETIC Sub-window

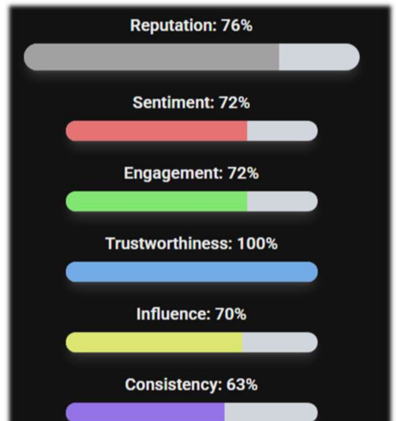


Figure 21. Web Application - SETIC Scorings

Where, for each SETIC component, on-toggle, it displays underneath a detailed contextualization and insights for the given profile – as seen in (S) [Figure 25.], (T) [Figure 22.], (C) [Figure 23.], (I) [Figure 24.], and (E) [Figure 26.].

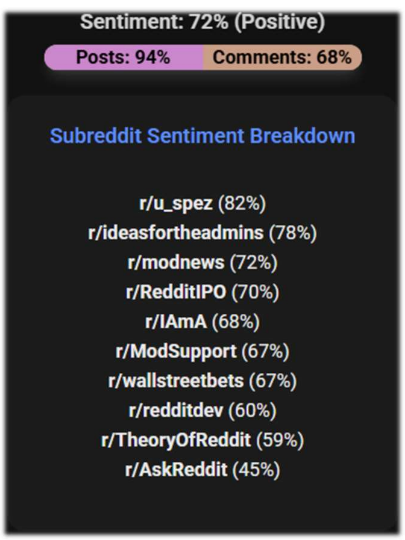


Figure 25. Web Application - (SETIC) Sentiment Scoring

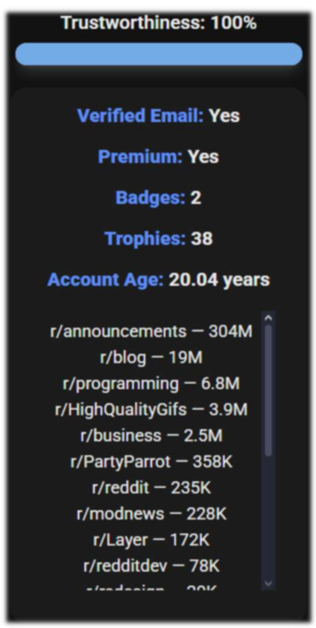


Figure 22. Web Application - (SETIC) Trustworthiness Scoring

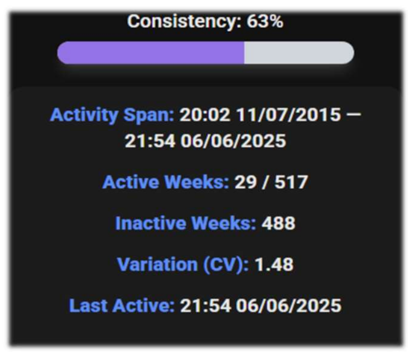


Figure 23. Web Application - (SETIC) Consistency Scoring

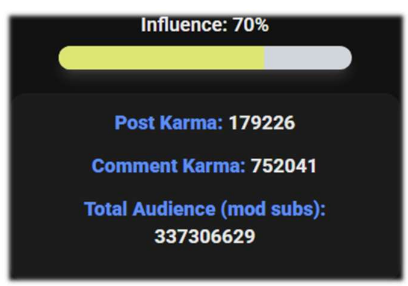


Figure 24. Web Application - (SETIC) Influence Scoring

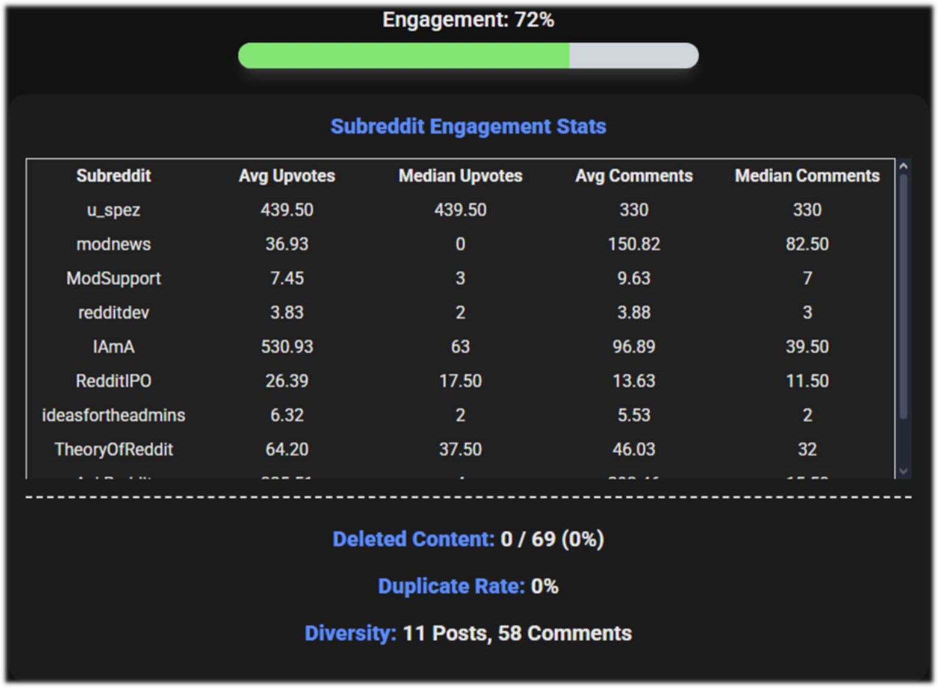


Figure 26. Web Application - (SETIC) Engagement Scoring

As seen in [Figure 14.], the web application goes beyond single-profile exploration, allowing users to toggle from the header a CompareMode state for any selection of sub-windows. Profiles submitted while this mode is active are automatically added to the graph datasets of the selected sub-windows (matching the datasets based on category). This mechanism allows social researchers or analysts to discern patterns, spot outliers, and track behavioral correlations between various cases. Despite the mergence of the datasets to result in comparative visualizations, each profile maintains its own dataset and SETIC score view. A user can simply toggle CompareMode off or on, to change the graph visualization from the combined dataset to their singular dataset and vice-versa, whereas the SETIC sub-window calculates the reputation and SETIC scores for only the primary profile (dataset), regardless of the state of CompareMode.

This modular, self-updating architecture empowers even the most inexperienced users with an easy-to-use interface. By supporting multi-profile analysis, real-time scoring, and dynamic visual exploration, SocialSentrix offers any user the ability to extract useful insights from complex social media data, regardless of whether they’re a research team, company department, or even just an average social media user. The platform’s overall objective is to present comparative reputation intelligence in an educational and user-centric way, backed by highly flexible and insightful data visualization tools.

5.4.4. State Management, Responsiveness, and Routing

The frontend routing architecture is structured using *react-router-dom*, with all page-level routes being defined in *App.jsx*. While individual page components, such as *Home.jsx*, *About.jsx*, *ManageSessions.jsx*, *Parse.jsx*, and *Redir.jsx* are dynamically loaded based on the user's navigation path; the *Header.jsx* and *Footer.jsx* components are globally rendered in the application's layout, across all frontend views. From a UI design and data flow standpoint, SocialSentrix uses React's *useState*, *useEffect*, and browser cookie handling to manage the state of user sessions, input fields, login status, multi-window interaction, and so much more. When a user loads the application, the global *Header* component checks for a valid *userID and sessionID* pair, based on which, it changes the elements from their default states and values. As profiles are submitted, the *ResultedWindow* component modularly incorporates, tracks, and conditionally renders the necessary subcomponents (*SortableResultWindow*, *SubWindow*, *ChartBlock*), preventing any unnecessary DOM updates. The responsiveness is achieved through these flexible component designs and modular state updating. While all the elements (such as input history, window position, popup visibility, etc.) are tracked locally, the reactive updates (based on the generated component tree), ensures that, even in the case of numerous component and subcomponent elements, behavioral predictability and high performance is still achieved, which is an essential factor when it comes to extensive researching and comparisons.

Additionally, another role of routing is to direct user exploration. For example, the */redir* path is used as a fallback, designed to gracefully return the user to the homepage ("/" route) in case of stray, undefined, or flawed URLs. The *Header* and *Footer* provide visible and constant access to the base homepage */* and */about* informational page, no matter where the user might be within the frontend of the web app. */parse* offers similarly to */redir*, a graceful method to visually display confirmations, alerts, and errors from certain backend requests. Even the */managesessions* route has its own role to offer a protected and cookie-validated UI for users to manage the sessions and requests tied to their account.

Furthermore, SocialSentrix uses a hybrid styling approach to maintain consistency and clarity, not only in layout and appearance, but in code writing as well. The entire application utilizes *Shadcn* UI components and Tailwind CSS for quick, uniform, responsive UI elements and utility-first styling. Most of the design-specific and reusable UI patterns are split into multiple CSS and JS files, allowing for easy, local importation across all components. Through this style modularity of component-specific CSS files and global JS interactive elements, we ensure a stable frontend environment where global styling conflicts are reduced, promoting effortless future development. Allowing the developer(s) to retain fine-grain control over component behavior and interactivity while maintaining a modern and consistent look.

Altogether, the frontend of SocialSentrix is built to be modular, resilient, and highly responsive to both user interaction and backend state changes. The application effectively combines technical complexity and usability with a strong architecture that manages routing, state management, asynchronous data fetching, authentication, and visualization, all within an easy-to-use interface. Thus, allowing both novice users and experienced analysts to effectively interact with multi-dimensional social media data and obtain insightful knowledge.

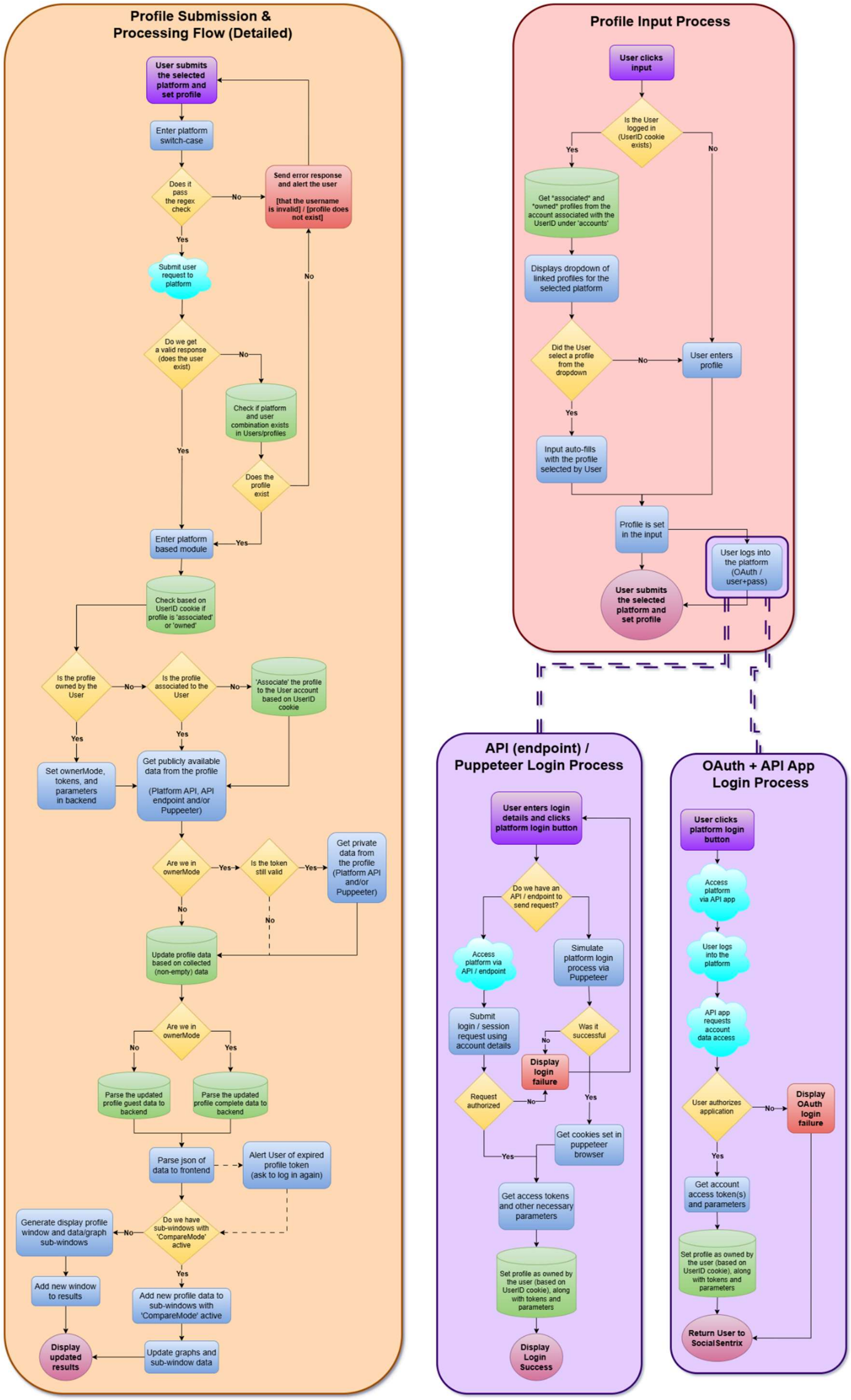


Figure 27. Profile Input, Submission, and Platform Account Login Conditional Logic

5.5. Backend Infrastructure and Crawling Logic

SocialSentrix's backend is built using a modular Express.js server architecture, containing platform-specific routing and controller logic (endpoints) assigned to specific files. The */routes*, */controllers*, and */services* directories contain the backend integration flow for Reddit, Bluesky, and Twitter (authentication, profile processing, and reputation score calculation), as well as the profile submission endpoint, MongoDB processing, and more. This separation enables a scalable development, consistent error handling, and customized scraping/authentication techniques based on each platform's API and public-facing interface.

5.5.1. API Structure, Middleware, and Platform Routing

The backend architecture of SocialSentrix is designed to allow both for internal frontend integration and direct, external API-level access. The initialization of the backend is performed by *server.js*, where it initially creates a MongoDB connection (through the *connectMongo()* function) and afterwards utilizes Express.JS to initialize the API server (*app = express())*. The express application instance enables routing definition (*frontendAcess.js* & *routing.js*), middleware configuration (*cookieParser* & *express.json*), HTTP request and response handling, as well as listening for incoming connections (*app.listen()*). The two middleware functions allow browser cookie manipulation and request body parsing, while the *frontendAccess* route module sets up the frontend connection based on whether the application is set in development or deployment mode (*setupDevCors(app)* or *serveFrontend(app)*). Afterwards, it runs the *startScheduledCleanup()* function to trigger the scheduled MongoDB cleaning, which removes all expired login and register requests every 6 hours. After all of the previous initializations and configurations, the API finally begins to listen for requests on the given port.

The root for all API endpoints is mounted at */api*, having r*outing.js* handle all of the sub-routing logic and endpoints, such as: */help* (lists the implemented paths), */test* (returns the status of the backend), */mongodb* (sub-router handling MongoDB based requests – */accountRegisterConfirm*, */loginStatus*, */manageSessions*, */getUserData*), */submit-profile* (handles the platform-specific profile submission), along with platform-specific sub-routers (*/reddit*, */bluesky*, */twitter*, */x*), and certain path catchers (*). Each platform-based sub-router (*reddit.js*, *bluesky.js*, *twitter.js*) has an Express router mounted at */reddit*, */bluesky*, and */twitter,* respectively, where they manage platform-based authentication (*/auth*), profile processing

(*/profile*), and reputation calculation (*/setic*), the flow of which can be seen in [Figure 27.]. Incoming traffic is routed to the relevant platform router by the main Express server (*server.js*, under the path */api*), using *router.post('reddit', redditProfile.get RedditProfile)* and other similar declarations. Each router calls upon the respective controller (*redditProfile.js*, *blueskySETIC.js*, *twitterAuth.js*, etc.), which handles the (endpoint-based) service requested (request validation, content extraction, token-based access control, puppeteer crawling, external API calling, etc.).

Due to the backend being built around loosely coupled components, it allows for flexible integration, testing, and reuse across various system components. This modularity also provides direct endpoint invocation and isolated debugging or dry runs. As shown in *apitest.js*, if a user desires to do so, they are able to make direct requests to the backend API, thus directly retrieving the data for the requested profile or the profile's reputation calculation. It also allows for platform authentication and profile (*guestView* / *owenerView*) data retrieval, based on the given *userID*. The supported requests and corresponding API responses include:

- Reddit profile submission: (POST) */api/reddit/profile*
  - Input: body {username}, header cookie {userID}
  - Response: json {platform, username, tokenInvalid, posts[], comments[]} and if profile owner (based on userID) {upvotes[], downvotes[], hidden[]} as well;
- Reddit profile auth: (GET) */api/reddit/auth*
  - Input: header cookie {userID}
  - Response: redirect {oauth2 redirect url};
- Reddit profile SETIC scoring: (GET) */api/reddit/setic*
  - Input: query {username, userID, startTime, endTime}, header cookie {userID}
  - Response: json {**S**:[score, label, avgPosts[], avgComments[], perSubreddit[], n], **E**:[score, perSubredditStats[], deletionStats[], duplicationStats[], diversityScore[]], **T**:[score, has_verified_email, is_gold, badges[], trophies[], ageYears, moderated Subs[]], **I**:[score, postKarma, commentKarma, totalMembers], **C**:[score, activity Span[], totalWeeks, activeWeeks, inactiveWeeks, cv, lastActive], **R**}
- Bluesky profile submission: (POST) */api/bluesky/profile*
  - Input: body {username}, header cookie {userID}
  - Response: json {platform, username, tokenInvalid, posts[], comments[], reposts[]} and if profile owner (based on userID) {likes[]} as well;

- Bluesky profile ATprotocol auth: (POST) */api/bluesky/auth*
  - Input: body {username, password}, header cookie {userID}
  - Response: json {success, handle}
- Bluesky profile SETIC scoring: (GET) */api/bluesky/setic*
  - Input: query {username, userID, startTime, endTime}, header cookie {userID}
  - Response: json {**S**:[score, label, avgPosts[], avgComments[], n], **E**:[score, engagementRatio, likes, replies, reposts, totalCount], **T**:[score, followerCount, followingCount, externalDomain], **I**:[score, contentCount, totalLikes, totalReposts, avgImpact, followerCount, followerScore, impactScore], **C**:[score, activitySpan[], totalWeeks, inactiveWeeks, cv, lastActive], **R**}
- Twitter profile puppeteer auth: (POST) */api/twitter/auth*
  - Input: body {username, password}, header cookie {userID}
  - Response: json {success, handle}

Additionally, to maintain coherence across this thesis, the following (previous) sub-chapters elaborate on particular backend subsystems:

- ✦ **3.3.1.** and **3.3.2.** describes the SETIC-based reputation scoring logic, mathematical models, and data usage process;
- ✦ **5.4.3.** discusses the platform-based authentication mechanisms, including OAuth2 / atproto api / puppeteer flow, token capture, and storage;
- ✦ **5.1.3.** outlined the profile submission flow, including platform detection, profile regex check and username normalization, backend dispatching, and more;

5.5.2. Reddit Profile Data Gathering

The *RedditProfile.js* controller handles the collection of Reddit-based profiles, initially associating the profile with the logged-in user's *userID*. Unlike Bluesky, Reddit has no dedicated API library or agent; however, it has two separate endpoints for accessing public and private data: "*https://api.reddit.com/user/<username>*" and "*https://oauth.reddit.com/ user/<username>"* for which we provide the header "*Authorization: `Bearer ${token}`, 'User-*

Agent': 'SocialSentrixBot/1.0 (by u/SocialSentrix)' ”. Based on the content type, there are two functions used to retrieve the data. For the profile’s metadata (*/about* and */trophies*), we use *safeFetch()*, whereas for the posts, comments, upvotes, downvotes, and hidden content (api.reddit: */submitted*, */comments* | ouath.reddit: */upvoted*, */downvoted*, */hidden*), we use *fetchAllItems*(). The difference between the two being that *safeFetch()* is used for single-page / non-paginated endpoints, while *fetchAllItems()* is used for paginated endpoints. Similarly to the *getAllPages()* function mentioned below in Bluesky’s case, *fetchAllItems()* repeatedly calls *safeFetch()* in order to get 100-item pages, stopping once it gets an empty result. Once the raw data is all collected, it is fragmented into different strings, objects and arrays, such as: *cakeDay* (*about.data.created_utc*), *karma{}* (*about.data.{comment/link/ total}_karma*), *about{}* (*about.data.{subreddit/is_gold/has_verified_email/icon_img/ snoovatar_img}*), *trophies{}*, *posts{}*, *comments{}*, *badges{}*, *profileName*, *description*, *moderatedSubs{}*, *upvoted{}*, *downvoted{}*, and *hidden{}*.

Additionally, to enrich the collected data beyond the received JSONs, through the *ScrapeRedditExtras()* function, Puppeteer is used to scrape auxiliary profile data such as: badges (“*faceplate-tracker*”), profile name (h2-based), description (“*[data -testid="profile-description"]*”), and moderated subreddits (“*ul[role= "menu"]*”). These scrapes are then decluttered in order to return the necessary values, which are then added to the *guestView* object. As mentioned previously, to avoid data redundancy/deletion, the content is then passed through the *buildMeaningful Update()* function, which only writes new or updated fields to MongoDB. Finally, the completed document is then returned as a result (the *ownerView* being sent only if *ownerMode* is set as true).

5.5.3. Bluesky Profile Data Gathering

The *BlueskyProfile.js* controller handles the collection of Bluesky-based profiles, initially associating the profile with the logged-in user’s *userID*. Then, based on the profile’s ownership and token validity, using a new *BskyAgent*, it resumes a session using the previously retrieved profile account’s decentralized identifier, access token, and refresh token (*did*, *accessJwt*, *refreshJwt*). This allows us to later get the profile’s likes; however, if the user does not own the profile, then we continue to fetch the publicly available data (through the official ATProto API), such as:

- Profile metadata (*agent.resolveHandle*, *agent.getProfile*);

- Posts (agent.getAuthorFeed, by filtering the *rawFeed* with ``` *p.post?.record?.$type === 'app.bsky.feed.post' && !p.reply && !p.reason* ```);
- Comments (agent.getAuthorFeed, by filtering the *rawFeed* with ``` *p.post?.record?.$type === 'app.bsky.feed.post' && p.reply* ```);
- Reposts (agent.getAuthorFeed, by filtering the *rawFeed* with ``` *p.reason?.$type === 'app.bsky.feed.defs#reasonRepost'* ```);
- Follower count (*agent.getProfile.data.followersCount*);
- Following count (*agent.getProfile.data.followersCount*);
- Likes [private data] (*agent.getActorLikes*);

In Bluesky's case, a *getAllPages* function is used to retrieve all the paginated data, by repeatedly calling a data fetching function (*fn* – based on the methods mentioned above) with a *cursor* (string/token sent from the ATProto API, pointing to the start of the next page), continuing until we get an empty page or the signal that there's no new page left (the API doesn't return a cursor). Furthermore, for consistency across all platforms, the gathered data for Bluesky is arranged into the two groups: *guestView* and (if applicable) *ownerView*. As mentioned in previous chapters, before saving/updating the profile in MongoDB, it is passed through the *buildMeaningful Update()* function, and afterwards, a final request is made towards MongoDB to request the complete and updated data, which is then returned as a result (the *ownerView* is sent only if *ownerMode* is set as true – the user owns the profile).

5.6. Configuration, MongoDB, Error Handling, Fallbacks, & Optimizations

SocialSentrix uses a meticulously planned backend architecture to ensure system dependability, scalability, and adaptability across different runtime environments. This includes a strong MongoDB (document-based) database system, extensive fallback procedures and error handling, the usage of environment variables for dynamic runtime configuration, and an organized folder structure dedicated to splitting the entire project, its components, and sub-components into a modular, tree-like structure. The following subchapters describe the backend architecture and techniques used to support and improve the previously mentioned capabilities.

5.6.1. Project Structure and Runtime Configuration

The overall project is split into two distinct directories: */client/* and */server/*. The client folder hosts the entire frontend React-based web application, including the frontend and entry point (*index.js)* and configuration files (*.env*, *components.json*, *package.json*, etc.), */assets/* (containing all of the custom visual content), */pages/* (route-based pages), */sections/* (global components – *Header* and *Footer*), */components/* (the main components – *SubmitProfile*, *ResultedWindow*, *RegistrationPopup*), */subcomponents/* (subcomponents – /*Registration/* and */ResultedData/*), */styles/* ((sub-)component-based stylesheets), */ui/* (reusable *Schadcn*-based UI components), routing logic (*App)*, and much more. Similarly, the server folder hosts the entire backend *Node.js*-based JavaScript applications, including the Express.JS logic, backend entry point (*/api* set in *server*.js) and configuration files (*.env*, *package.json*, etc.), */routes/* (platform-based routes, *frontendAccess*, *submitProfile*), */services/* (containing MongoDB connection and interaction logic), */controllers/* (platform-based authentication, profile processing, SETIC score calculation; account registration, session handling, data retrieval, etc.), and more. This project segmentation provides a effortless logic flow, ensuring a clean separation between layers and the used components.

Both frontend and backend heavily utilize environment variables (defined in their respective *.env* files) in order to ensure deployment flexibility and allow external service changes, by hosting variables such as: The MongoDB URI, database, and collection names, Reddit App configuration, SMTP email credentials, and more. By simply changing the relevant environment variables, the project can easily transition between various environments, including development (localhost), staging (internal IP), and production (Google Cloud / External IP), without requiring heavy code changes throughout the application.

5.6.2. MongoDB Collections and Data Organization

MongoDB serves as the main data storage system, thanks to its adaptable document-based structure, which is ideal for hosting, updating, and retrieving large amounts of dynamic profile data, as well as accommodating distinct platform-specific schemas. This flexibility allows SocialSentrix to host all its persistent data within a singular database on the designated *SocialSentrixCluster*. When a user submits a profile for analysis, the backend parses the collected data through a *buildMeaningfulUpdate()* helper function. This function minimizes unnecessary write operations by comparing existing and incoming fields within the selected documents, ensuring that unchanged or empty values do not overwrite existing data. The *Users* database houses four main collections:

- *accounts* – Stores SocialSentrix accounts, where each document representing a user includes 6 fields:
    - id: A unique UUIDv4 userID;
    - email: The user's email address;
    - creationDate: A timestamp of when their account was confirmed;
    - adminSessions: An array of session IDs granted administrative privileges;
    - associatedProfiles: An array of objects, each representing a previously processed profile, containing the profile's platform and username;
    - ownedProfiles: An array of objects, each representing a previously logged-in platform account, containing the account's platform, username, and authentication token(s) (reddit_token / did, accessJwt, refreshJwt / auth_token, ct0);

- *pending* – Stores the account registrations which are awaiting confirmation, each document hosting three data fields:
    - token: A UUIDv4 token used for confirming the creation of the account;
    - email: The user's email address;
    - createdAt: A timestamp of the initial registration request;

- *profiles* – Stores all the user profiles, regardless of whether they are linked to a SocialSentrix account, each document hosting three to four data fields:
    - platform: The name of the profile's platform;
    - username: The profile's username;
    - guestView: Publicly available data collected from the profile;
    - ownerView (optional): Privately collected data available only after platform authentication/authorization;

- *requests* – Stores all the requests made by users attempting to log into a SocialSentrix account, each document hosting ten fields:
    - sessionID: The session identifier of the user that made the login request;
    - email: The target account's email address;
    - name : The name given with the request;
    - comment: An optional comment given with the request;

- userAgent: Details about the requester's browser and operating system;
- ip: The requester's IP address;
- location: An estimated geographic location of the requester;
- status: The current status of the request (pending/confirmed/denied/expired);
- createdAt: A timestamp of when the request was created;
- expiresAt: A timestamp of when the request will expire;

5.6.3. Error Handling and Fallback Mechanisms

SocialSentrix also implements an extensive fallback logic and error handling system, where not only are there frontend and backend stray routes/endpoints handlers (*/redir*, */parse*, */api*), but there is also a substantial usage of try-catch blocks. For example, when a profile does not exist within a platform or MongoDB, a 404 error code is returned, and an alert message is rendered under the "Submit Profile" button. When a platform account token is invalid or expired, a 401 error code is returned, and an alert appears notifying the user of this issue. For all email confirmations, if an issue is encountered in the backend, the user is then redirected to the frontend */parse* route, visually displaying the issue met (such as the account being already confirmed, a request being expired, etc.). For all backend requests, there is a catch-all added for any unforeseen issue or error, returning a 500 error code. These are only a few of the multitude of cases where a fallback logic or error handling system intervenes.

5.6.4. General Optimization Techniques

Additionally, several optimization strategies have been implemented throughout the application to improve performance, reduce redundancy, and ensure scalable data handling. These include: the previously mentioned concurrent profile submission requests (through *Promise.all()*) in order to parallelize data fetching and crawling; token-aware crawling; the usage of update operators ($set, $push, $unset, etc) in MongoDB requests; customized API requests, where the requests are done in limited batches with break time intervals; the usage of additional mechanisms and plugins for Puppeteer, such as *stealth*, *--no-sandbox* arguments, custom headers, waits and timeouts to simulate human-like behavior (through delays, scrolls, interactions, etc.); and so many more.

Together, all these strategies help the web application maintain responsiveness and stability while also providing robustness and fault-tolerance.

6. Conclusion & Future Work

This chapter reflects upon the outcomes of the research and development of the SocialSentrix application, assessing whether the initial goals were met, describing the constraints faced during development, as well as future plans for growing the platform.

6.1. Summary of Research Contributions

This dissertation introduced SocialSentrix, a full-stack web application for evaluating and determining social media profile reputation scores via the proposed SETIC framework. The system employs a modular architecture, allowing users to view, compare, and analyze multi-dimensional activity data in real time through the integration of platform-specific data collectors. Going beyond metrics and visualizations, it was also able to calculate one's overall reputation score through the use of the SETIC framework, comprising of individual Sentiment, Engagement, Trustworthiness, Influence, and Consistency scores. Throughout development and research, it was able to complete and even go far beyond the initial objectives mentioned in chapter **1.4.**. Profile data collection, platform authentication, and SETIC calculation were successfully implemented for two highly popular and vastly different social media platforms (Reddit & Bluesky). The application emphasizes both modularity and reusability by separating and organizing the codebase and its data structures based on platform logic, user session management, analytical rendering, and more, allowing robust testing, maintainability, and extensibility throughout the project's development. This prototype demonstrated the importance and technical validity of such a tool for academic, personal, business, and scientific purposes. While the application does indeed host processing for only 2 platforms, it lays the foundation for seamless future growth and improvement. Thus, this academic paper and it's codebase lay the groundwork for upcoming tools in reputation monitoring, digital identity auditing, and comparative social media behavior analysis.

6.2. Limitations, Challenges and Alternative Approaches

Despite its achievements, the platform is not without limitations. All platforms have noticeably presented certain access restrictions and technical constraints, such as API rate limits, data retrieval restrictions, request throttling, denial of headless browser approaches, general fragility, and so much more. While several of these challenges were mitigated through solutions such as code pauses, request timeouts, user interaction simulation, repeated requests, and auxiliary plugins, many other solutions and optimizations have not yet been possible. For instance, the SETIC scoring model is limited by the quality, granularity, and availability of accessible data. One proposed optimization was in regards to the Engagement calculation, to evaluate a profile's content based on the activity levels of all content within the subreddit during a two-week window before and after the original profile's submission, thus taking into account possible subreddit activity fluctuations. This proposal was not possible due to recent Reddit API constraints and the infeasibility of indexing all of the subreddit's content. Another limitation lies in the lack of deeper profile data access and server logs. If such access were provided, it would be possible to collect elements such as reports, shadow bans, and multi-account usage (based on the IP and hardware), which would greatly influence a profile's trustworthiness score. Additionally, there are also technological limitations, such as for sentiment analysis. While the VADER model is efficient for short-form content, it may underperform in cases of mass, nuanced text processing, in which case Apache Spark and HDFS can be used to drastically reduce the overall computation time. These examples represent only a subset of the challenges met and opportunities possible, clearly underlining the possibility of drastic and effortless (due to its modularity) enhancements, thus providing more metrics and more precise calculations.

6.3. Future Development and Expansion

SocialSentrix was designed with extensibility in mind, both in terms of technical architecture and analytical capability. Depending on future data, platform, and technological access, some later development efforts may include:

- *Support for new platforms* – Including analysis modules for other popular social media platforms such as LinkedIn, Instagram, Mastodon, etc;
- *Multimedia processing* – The implementation of media-aware sentiment scoring extensions based on machine learning for image, audio, and video content;
- *Mass comparative analytics* – Allowing bulk profile ingestion by geographic area, community, groups, etc;
- *WebIDMass comparative analytics* – Allowing bulk profile ingestion by geographic area, community, groups, etc;

- *Increasing platform rate limits* – Being able to perform more actions/requests than the set platform rate limits (Reddit: 100 queries per minute per OAuth client | Bluesky: 3000 requests per 5 minutes / 30 session creations per 5 minutes and 300 per day, per account);
- *Session administration expansion* – Expanding the administrative UI and tools to allow for session grouping, automatic request acceptance and denials, etc;
- *SETIC scoring evolution* – Saving a profile's SETIC scores besides its data that changes can be visualized and tracked;
- *User-Type adaptive weighting* – Changing the SETIC component weights and calculation methods based on the user type (business accounts, lurkers, moderators, casual users, content creators, etc.)
- *Profile social graph* – Creating a profile's social network based on visible user interactions (comments, reposts, follows) as another contributor when calculating one's trustworthiness score;
- *Further data inclusions and statistics* – Such as comment statistics (number of comments, upvotes, and reposts), post view count, Reddit content award data (number of awards and their costs);
- *Moderation weight changes* – Taking into consideration profile moderation status when calculating an engagement score, so that they are not artificially inflated due to account status;
- *Topical consistency* – Clustering profile's content, checking if they are thematically consistent;
- *SETIC weight adaptation* – Changing the component weights based on the respective platform;
- *Advanced NLP systems* – Adding NLP systems like BERT to enhance sentiment scoring;
- *Scaled Bluesky virality* – Developing scaled models for viral content, adjusting based on the account's overall statistics;
- *SETIC component interconnections* – Combining certain elements of the components to raise score accuracy due to their interlacing nature;

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AI Tool Usage Disclaimer

Throughout the development of this thesis, various artificial intelligence tools and non-AI tools were utilized to assist in the overall research and general writing process. These tools were carefully selected to assist with tasks such as idea/paragraph structuring, language refinement/formalization, resource and reference crawling, technical development, and general formatting improvements. Similar to non-AI applications, these tools were strictly supportive, where no particular AI tool contributed to the original research, critical analysis, or independent conclusions. All content was thoroughly conceived, reviewed, and compiled by ***Stan Andrei - Vlăduț***.